# Farshore

## Phase 3 UI design and usability report

CMPS 4200 HCI: AI Enhanced UI UX | Fall 2026
Bibek Pathak | Solo project | Prepared November 5, 2026

Five participants tested Cyclops' Cave. Entering the trial and recovering from an AI preparation failure were generally straightforward. Understanding the AI during fast play was harder: four participants found its information dense or hard to read. The most severe issues are now fixed in the game, with before-and-after screens in Section 4.

| Measure | Result |
|---|---|
| Mean SUS | 59.5 / 100, across 5 complete questionnaires (the Phase 5 baseline) |
| Mean AI trust | 3.0 / 5 |
| AI-panel ease | 2.8 / 5, across 5 recorded ratings |
| Recovery | 5 of 5 recovered without help; median 8 seconds |
| Data coverage | 18 of 20 task outcomes recorded. P1 T2/T3 outcome and time are missing. |
| Fixes from testing | B1-B4 and X1 built into the game and verified in the browser on November 5, 2026 |

### Deliverables and links

Raw data: Farshore-Phase-3-Test-Data.xlsx
Presentation: Farshore-Phase-3-Slides.pptx
Repository: https://github.com/RavangDai/Farshore
Screen design images: submitted as a separate zip file

Data note: for P2-P5 the moderator confirmed every measured value (task times, errors, ease ratings, SUS answers and trust ratings). Their free-text notes were reworded while the records were typed up, so this report uses them only as paraphrased notes, never as quotes (Section 3).

Contents: 1 Screen designs; 2 Test script; 3 Test report; 4 Changes and remaining work; 5 AI tool use; 6 Team contribution statement. Section 1 documents the design as it was tested; the study covers Cyclops' Cave only.

# 1. Screen designs

High-fidelity screens of the tested prototype, captured from the running game and annotated. The same frames are submitted as image files with this report. Numbered markers match the notes beside each screen: gold = interface, cyan = AI behavior or AI state. Failure states were produced with the mock AI service (Section 1.2), and the interface shown is the real one.

Flows: F1 Start, F2 Choose a trial, F3 Prepare the rival (the AI planning states), F4 Play, F5 Understand the rival (explainable AI), F6 Settings, F7 Account. Personas are the Phase 1 sketches: Maya (gamer, fairness), Daniel (reader, limited technical comfort), Rosa (AI skeptic). What changed from the Phase 2 wireframes, and why, is in Section 4, Part A.

## 1.1 Style basics (design system)

Decided before the screens and reused on every one. The values below are read from `src/styles/tokens.css`, the single file where the app defines its colours, type and spacing, when this document is built. Every stylesheet reads these variables.

### Colour palette

**Background** `--fs-color-bg`
#071523
Page and stage background

**Surface** `--fs-color-surface`
#0b2033
Panels and cards

**Text** `--fs-color-text`
#f6f0e3 · 16.2:1 on background
Primary text

**Muted text** `--fs-color-text-muted`
#93afc0 · 8.0:1 on background
Secondary lines and status text

**Accent** `--fs-color-accent`
#e9b85c · 10.1:1 on background
The one accent: buttons, focus, interactive chrome

**AI** `--fs-color-ai`
#32d4f5 · 10.4:1 on background
Only for AI state and AI data, never for hazards

**Error** `--fs-color-error`
#f4604f · 5.8:1 on background
Errors and real threats

**Success** `--fs-color-success`
#9fe6b4 · 12.7:1 on background
Confirmation and successful escapes

One accent, gold, for everything a player can press. Cyan is reserved for the AI, so "the rival is thinking" is never confused with "you are in danger" (red). Contrast ratios use the WCAG formula against the background. A few extra navy, gold and bronze tints are used only for depth and highlights.

### Type: four families, five sizes

| Family | Stack | Specimen | Use |
|---|---|---|---|
| **Display** `--fs-font-display` | Georgia, "Palatino Linotype", serif | Eight Trials. One Journey Home. | Story, headings, the rival panel |
| **UI** `--fs-font-ui` | Arial, Helvetica, sans-serif | Eight Trials. One Journey Home. | Small interface details |
| **Pixel** `--fs-font-pixel` | Pixel, monospace | Eight Trials. One Journey Home. | Arcade labels and eyebrows |

| Family | Stack | Specimen | Use |
|---|---|---|---|
| **Mono** `--fs-font-mono` | ui-monospace, SFMono-Regular, Consolas, monospace | Eight Trials. One Journey Home. | Inside the AI inspector only |

| Size | Value | Specimen | Use |
|---|---|---|---|
| **XS** `--fs-text-xs` | 12px | The rival is studying you | Captions, status lines, labels |
| **S** `--fs-text-sm` | 14px | The rival is studying you | Body text and buttons |
| **M** `--fs-text-md` | 16px | The rival is studying you | Panel titles, quotes, lead text |
| **L** `--fs-text-lg` | 24px | The rival is studying you | Section headings |
| **XL** `--fs-text-xl` | clamp(40px, 6vw, 72px) | The rival is studying you | Screen titles (scales with the window) |

### Spacing scale

| | |
|---|---|
| `--fs-space-1` | 4px |
| `--fs-space-2` | 8px |
| `--fs-space-3` | 12px |
| `--fs-space-4` | 16px |
| `--fs-space-5` | 24px |
| `--fs-space-6` | 32px |
| `--fs-space-7` | 48px |

In code: `tokens.css` defines the values, and the older colour names in `farshore-stage.css` are aliases of them. The AI rival panel, in all five states, uses only tokens. In the older stylesheets, every value equal to a token was replaced with that token. Off-scale values remain in in-game HUD labels and some menu rules, and are listed for Phase 4 (`docs/UI_DESIGN.md`).

## 1.2 AI components: five states and user control

The main AI feature is the rival's strategy. Before each round, the model reads the player's recent attempts and chooses how the mythic rival will play. It appears in the briefing's rival panel (flow F3) and is designed in all five states. The explainable AI Insights panel in Cyclops' Cave has its own empty state (D23).

| State | What the player sees | Frames |
|---|---|---|
| **1. Empty / initial** | The briefing opens before any strategy exists. Preparation starts at once, so the panel opens in the loading design at 0 s rather than as an empty box. With no model connected, the initial state is "No AI rival connected", with three ways to connect and Check again. In Cyclops' Cave, AI Insights reads "Observing. No committed action." until the rival's first decision. | D06, D10, D23 |
| **2. Loading** | "[Rival] is studying you", an elapsed-seconds counter and a cyan scanning bar, which does not pretend to know the remaining time. Matched to measured latency: the Phase 2 spike measured 7.3-11.6 s (mean 8.3 s) per strategy, and a CPU laptop with Qwen3-4B takes about 15 s. The mock's slow mode (12 s) tests this. Cancel, or Keep the previous strategy if there is one. | D06, D25 |
| **3. Success** | The strategy name, "Strategy chosen by [model]", a plain description, the rival's taunt (with Hear it), and Ask for another strategy. Start is enabled. | D07, D13 |
| **4. Low confidence** | "The rival's plan was set aside". The model's strategy did not pass the game's checks, so it was not used and nothing changed. The main action is Ask for another strategy. Keep the previous strategy appears if one exists. Amber, not red: nothing is broken. | D22 |
| **5. Error, with recovery** | "The rival is not ready", with a plain message and no codes. One clear action, Prepare the rival again, plus "Nothing about your voyage changes". Keep the previous strategy appears if one exists. Task 4 of the test leads participants here. | D09 |

### User control over AI results

| Control | How it works | Frames |
|---|---|---|
| **Regenerate** | "Ask for another strategy" on a ready strategy, and as the main action in the low-confidence state. The model chooses again. | D07, D24 |
| **Undo** | After asking again, "Undo: back to [previous strategy]" restores it, and pressing Undo again swaps back. While a new strategy is loading, or if it fails or is set aside, "Keep the previous strategy" returns to the old one. | D24, D25, D22 |
| **Reject** | The game rejects a strategy that fails its checks before it can be used (low-confidence state). Players can reject the AI commentary by switching commentary or voice off in Settings, or with the Voice button on the menu. | D22, D18 |
| **Cancel** | "Cancel preparation" stops a request in progress. The pilot found that the briefing then starts a new request at once. This is noted for the sessions. | D06 |
| **Edit** | Not offered by design: the strategy is the opponent's move. Changing the difficulty asks the model for a new strategy at the new level. | D08 |

### Mock AI service

All AI calls (rival status, strategy and live commentary) go through one module, `server/lib/model-api.ts`. It asks `server/lib/mock-ai.ts` first. The mock answers in the modes success, slow, uncertain, error and offline, plus "-once" modes that fail once and then recover. A moderator switches modes from a second browser tab at `/api/ai-mock?mode=...`, or for the whole server with `FARSHORE_AI_MOCK`. Farshore's live model is already connected (change A2), so in Phase 4 the mock stays as the test fixture for these states instead of being replaced. Frames marked "mock" were captured in the real interface with the mock answering, and name the model as "mock-ai (test mode)".

## 1.3 Desktop screens (1440 × 900)

### D01 Splash

F1 Start · All players




1. One call to action: press Enter or click to start. Keyboard, mouse and touch are all named.
2. Sign in is optional. Guests save on this device, as the top bar says.

## D02 Title menu

F1 Start · Daniel, Maya

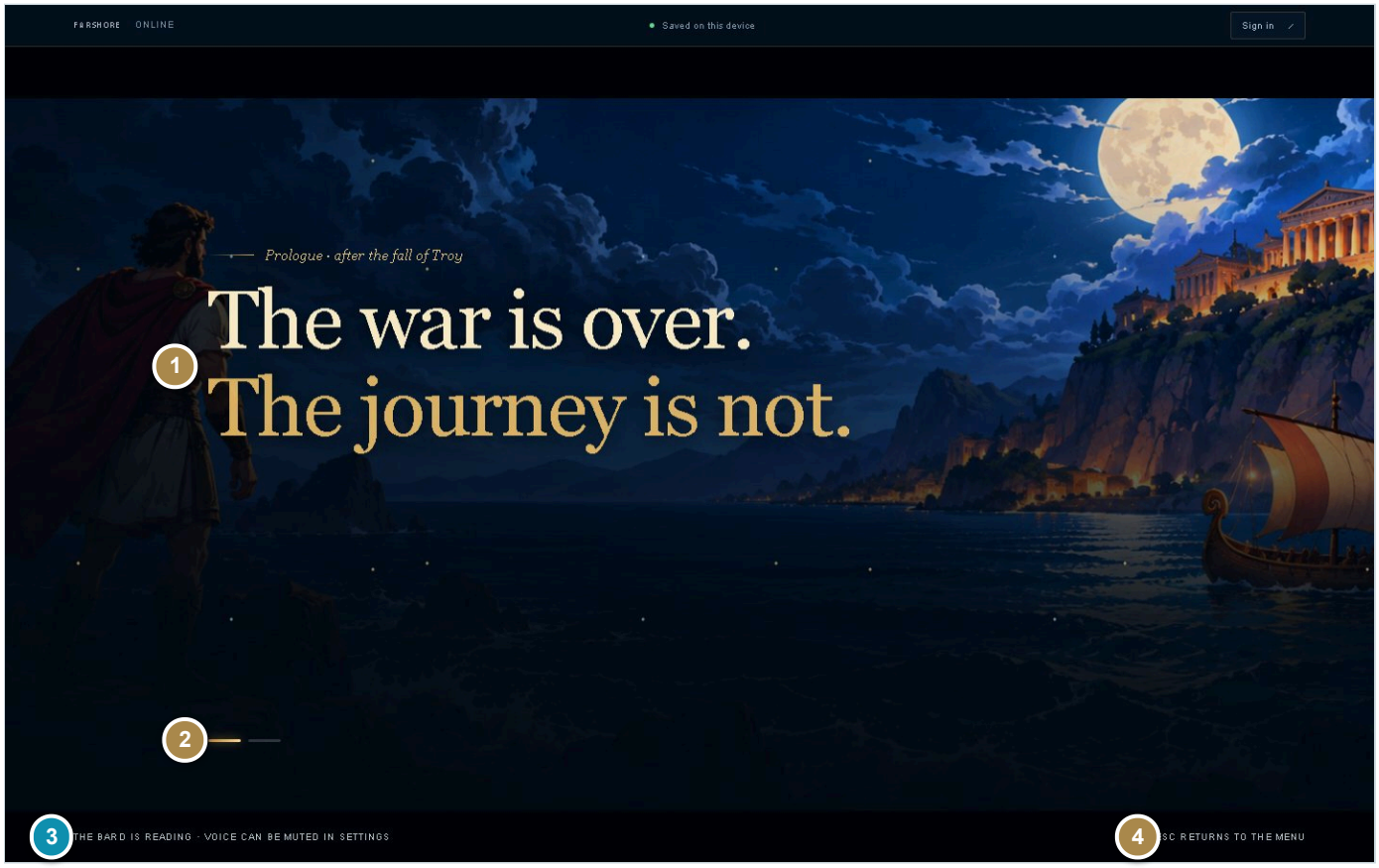



**AI state:** Rival status chip: checking, online (with model name) or offline (with Connect)

1. New game is the highlighted primary action. Every entry has a one-line explanation (T1).
2. Continue explains why it is unavailable: "No voyage saved yet".
3. The AI rival's status is visible before play, so a missing model is never a surprise mid-game (T4).
4. Voice and sound toggles on the first screen, without opening Settings.
5. AI Lab: watch the rival's decision pipeline without affecting the campaign.

## D03 Prologue

F1 Start · Daniel

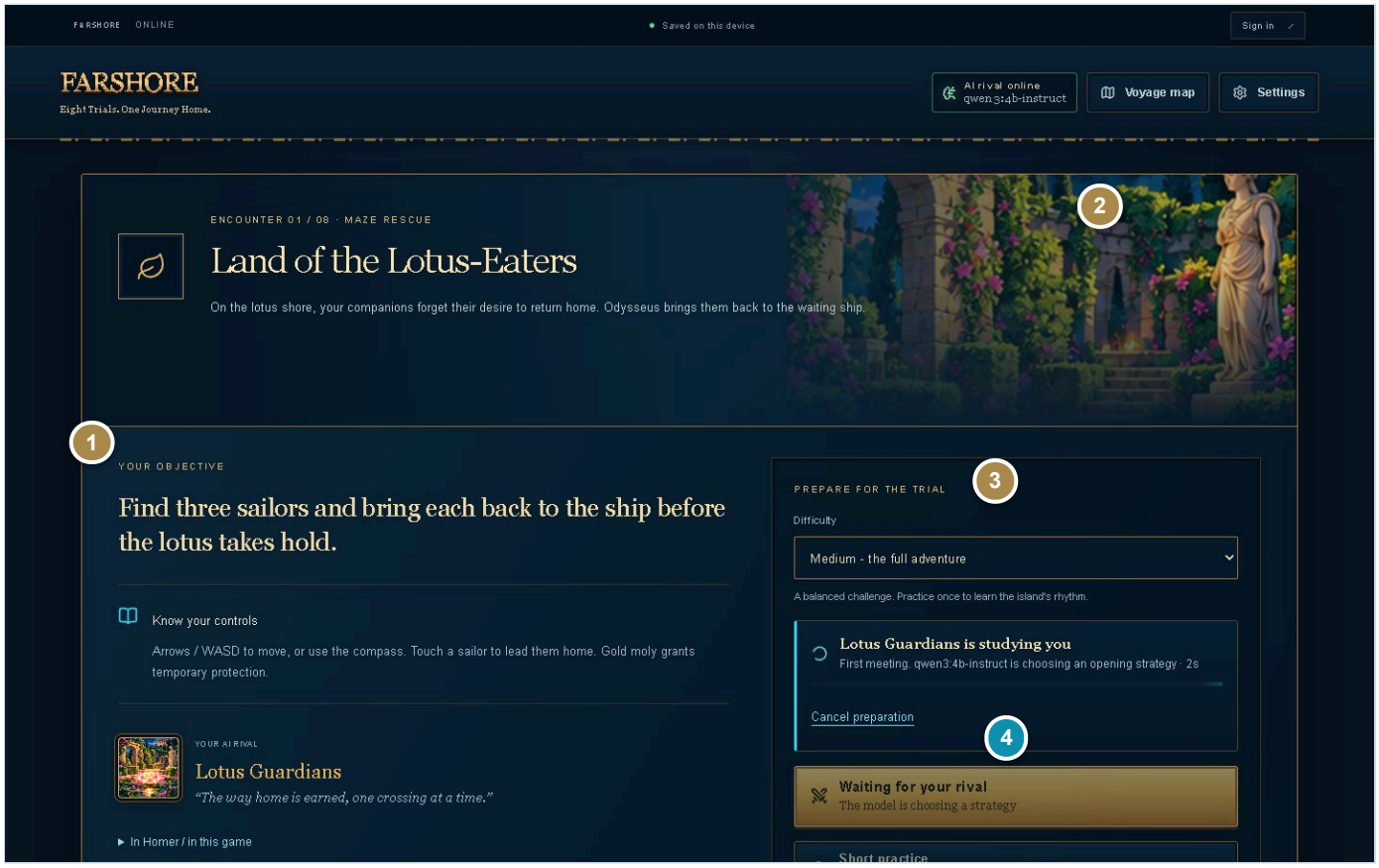



**AI state:** Bard voice (Kokoro-82M) reads each card aloud

1. Two short cards set the story. Large type, one idea per card.
2. Progress dots show how many cards remain.
3. States that the bard is reading and where to mute the voice.
4. Escape returns to the menu. The prologue can also be skipped to the voyage map.

## D04 Trial selection

F2 Choose a trial · Maya, Daniel

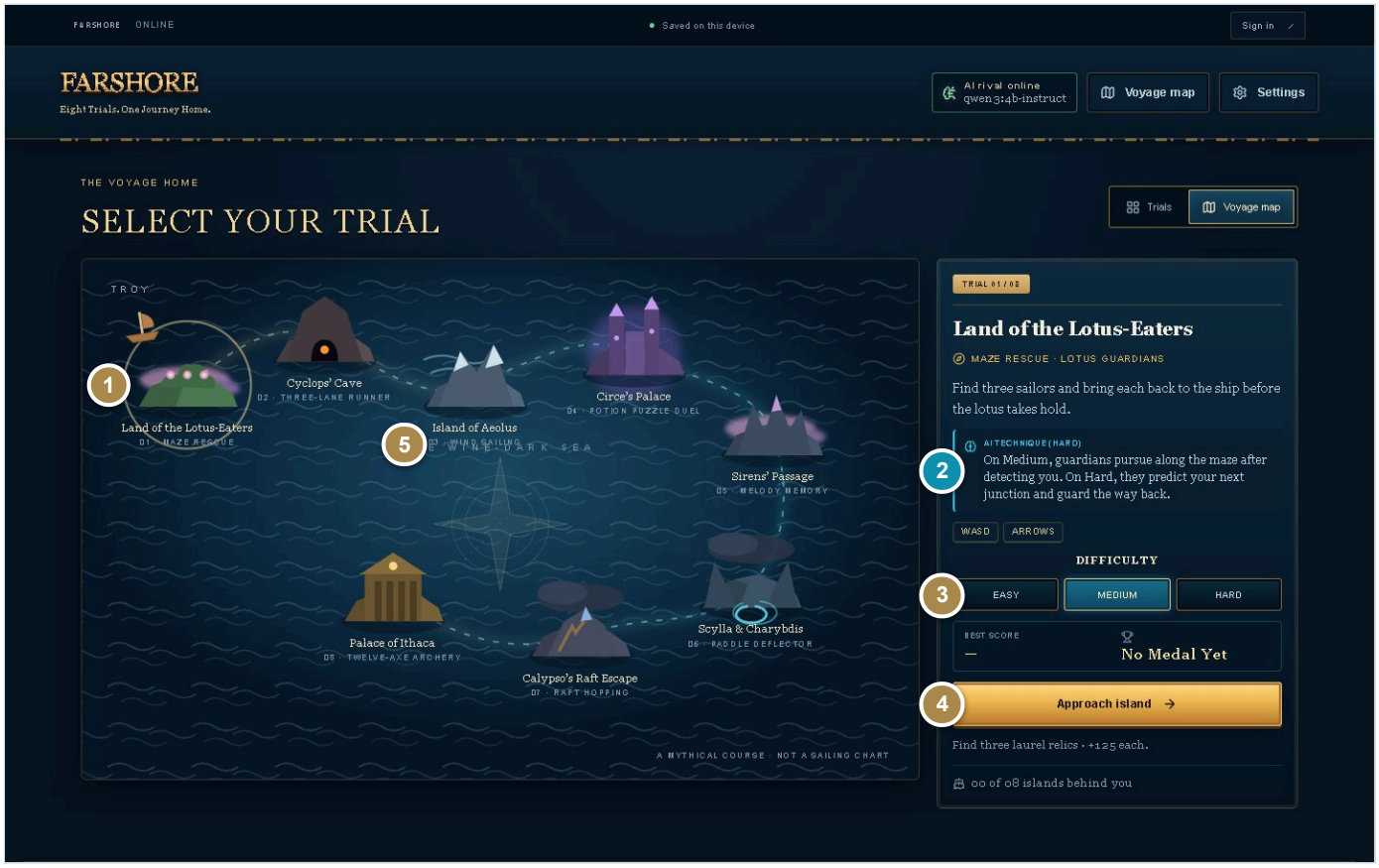



**AI state:** Per-trial note on the AI technique the rival uses at each difficulty

1. Eight numbered cards in voyage order. The selected trial has a gold outline.
2. Switch between the trial gallery and the illustrated voyage chart.
3. The side briefing previews the objective before the player commits.
4. Explains how the rival behaves on Medium and Hard, in plain words (transparency for Rosa).

## D05 Voyage chart

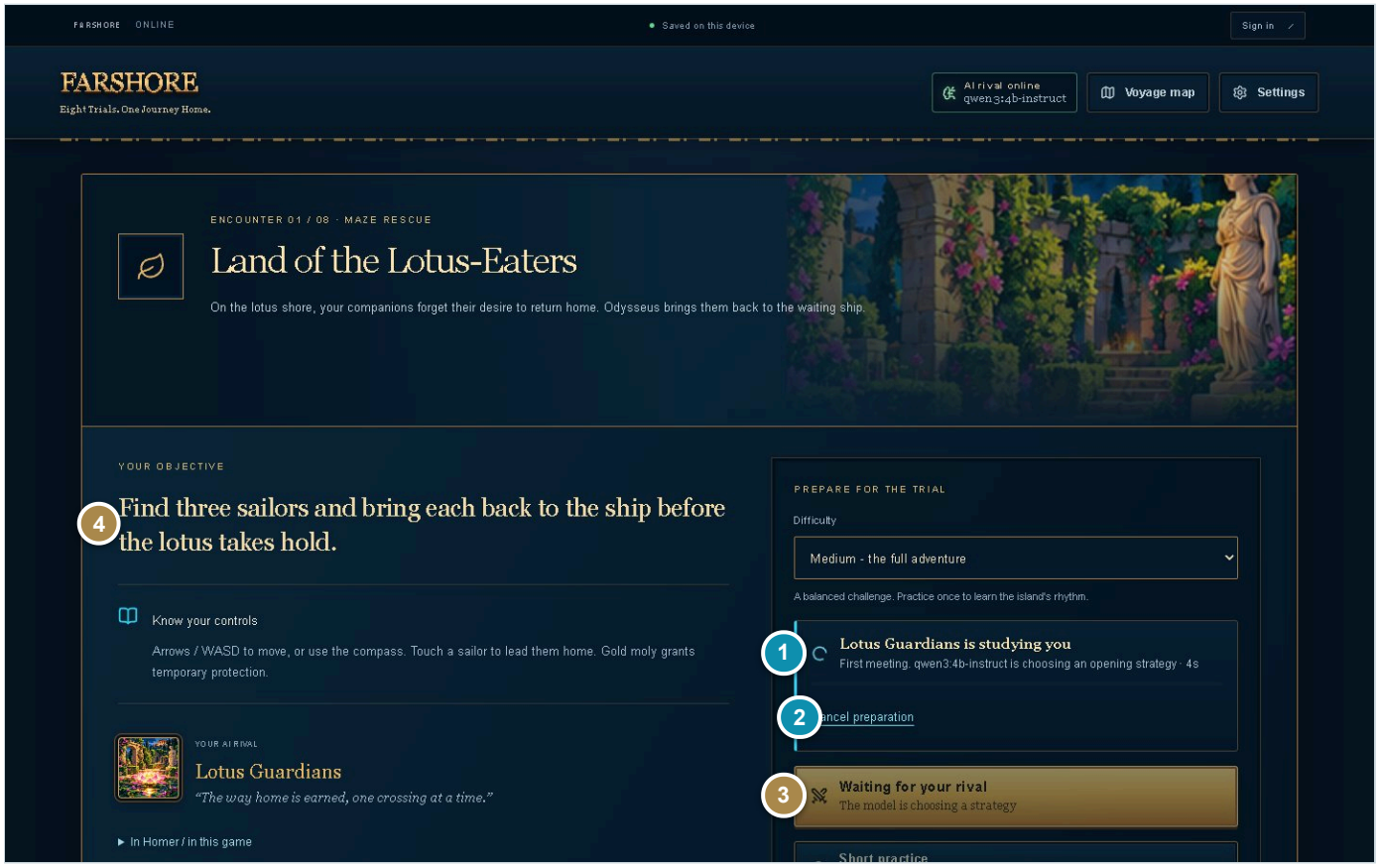
F2 Choose a trial · Daniel




1. The current island is ringed, and the dashed route shows the voyage order from Troy to Ithaca.
2. The same AI technique note as the gallery, beside the chart.
3. Difficulty can also be chosen here.
4. Approach island opens the briefing.
5. Pilot: some island labels overlap the "wine-dark sea" caption.
6. Locked, open and cleared islands differ by icon and label, not colour alone.

## D06 Briefing: the rival is preparing (states 1-2 of 5: initial and loading)

F3 Prepare the rival (A1 loading) · Rosa, Maya

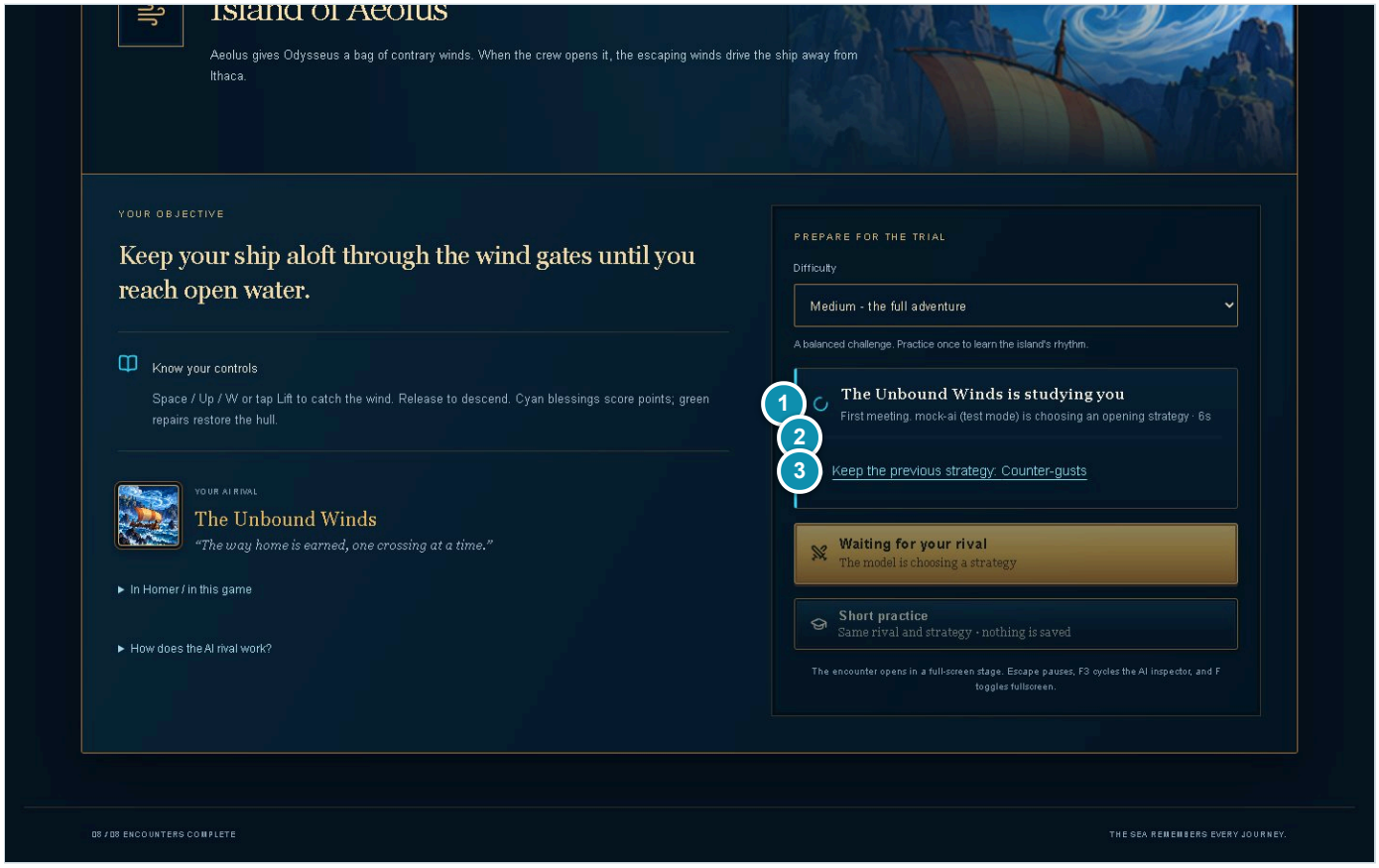



**AI state:** A1 loading: the model reads recent attempts and chooses a strategy (about 15 s on a CPU laptop)

1. Says what the model is doing ("is studying you"), names the model, says whether it is a first meeting, and counts the elapsed seconds.
2. Cancel preparation: the player is never stuck waiting.
3. Start says "Waiting for your rival" and is disabled until a plan is ready.
4. The objective and controls can be read while the model works, so the wait is useful.

## D25 Briefing: slow response while asking again (state 2 of 5: loading)

F3 Prepare the rival (A1 loading) · Rosa, Daniel

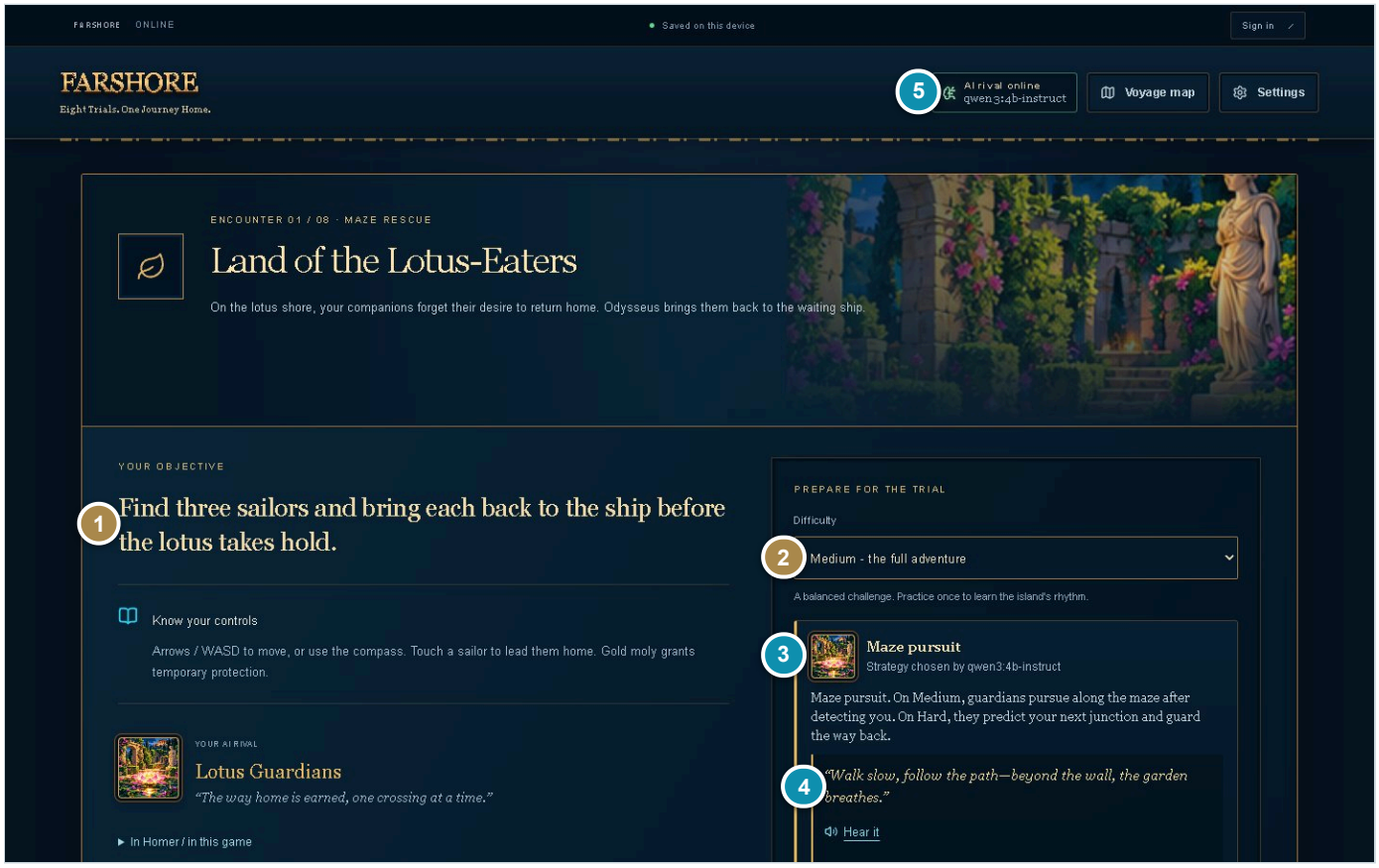



**AI state:** Loading during a slow response. Mock AI service in mode slow (12 s, just above the slowest strategy measured in the Phase 2 spike, 11.6 s)

1. Elapsed seconds instead of a fake percentage: the model's time varies, so the bar only shows activity.
2. Cyan scanning bar: cyan is used only for AI activity.
3. While a new strategy loads, the previous one is one click away (undo), so a slow model never blocks play.
4. Captured in the real interface with the mock AI service answering, so the model is named "mock-ai (test mode)".

## D07 Briefing: strategy ready (state 3 of 5: success)

F3 Prepare the rival (A1 success) · Rosa, Maya

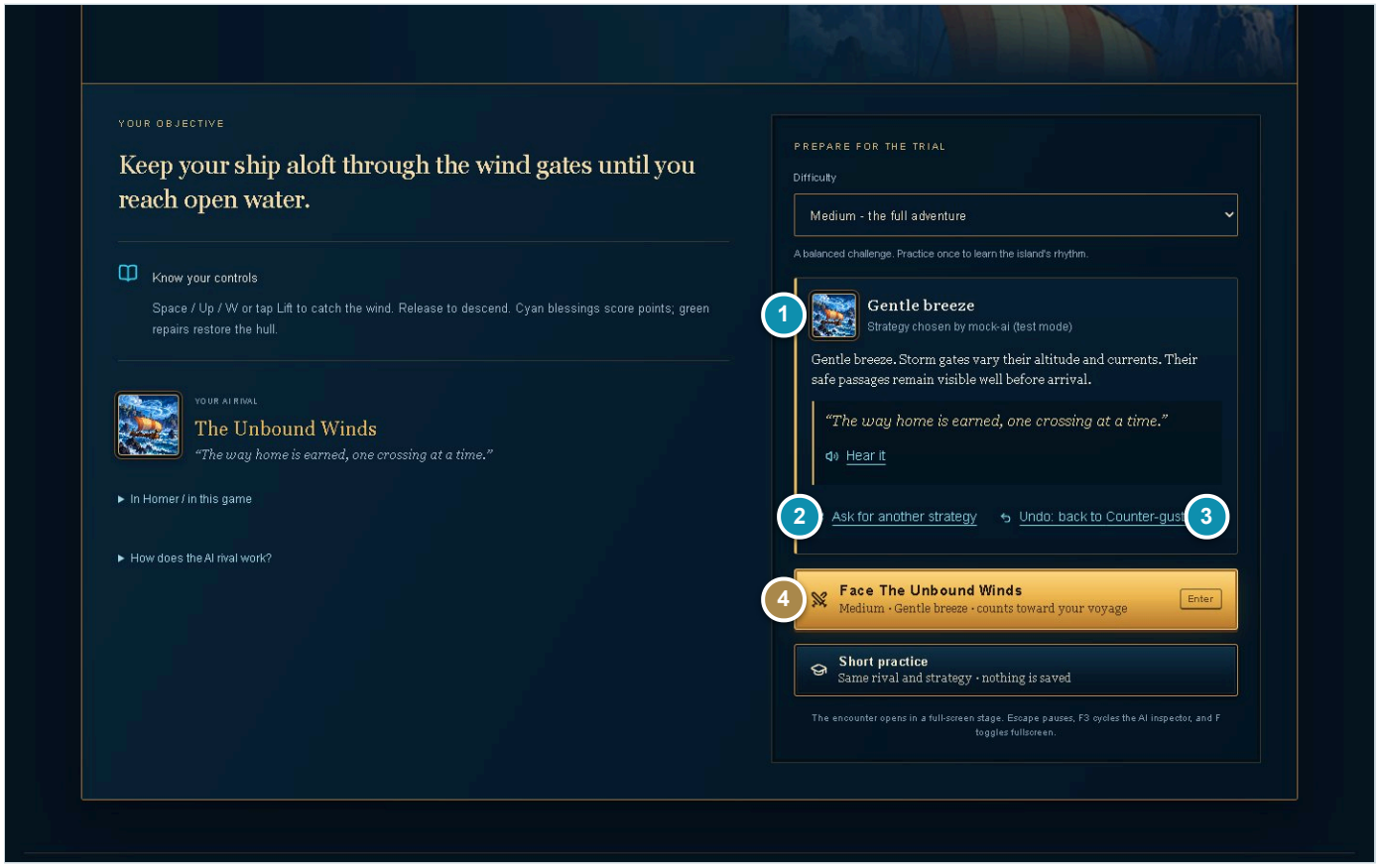



**AI state:** A1 success: named strategy, who chose it, description and taunt

1. The objective comes first, in plain words.
2. Difficulty with a plain-language note. Pilot: it defaults to Medium, which new players may not notice.
3. Strategy name and "Strategy chosen by qwen3:4b-instruct": the AI's decision is labelled as the AI's (T2).
4. Model-written taunt, with Hear it to play it in the rival's voice.
5. The rival status chip stays visible on every menu screen.

## D24 Briefing: another strategy, with undo (user control)

F3 Prepare the rival (A1 success, regenerate and undo) · Maya, Rosa

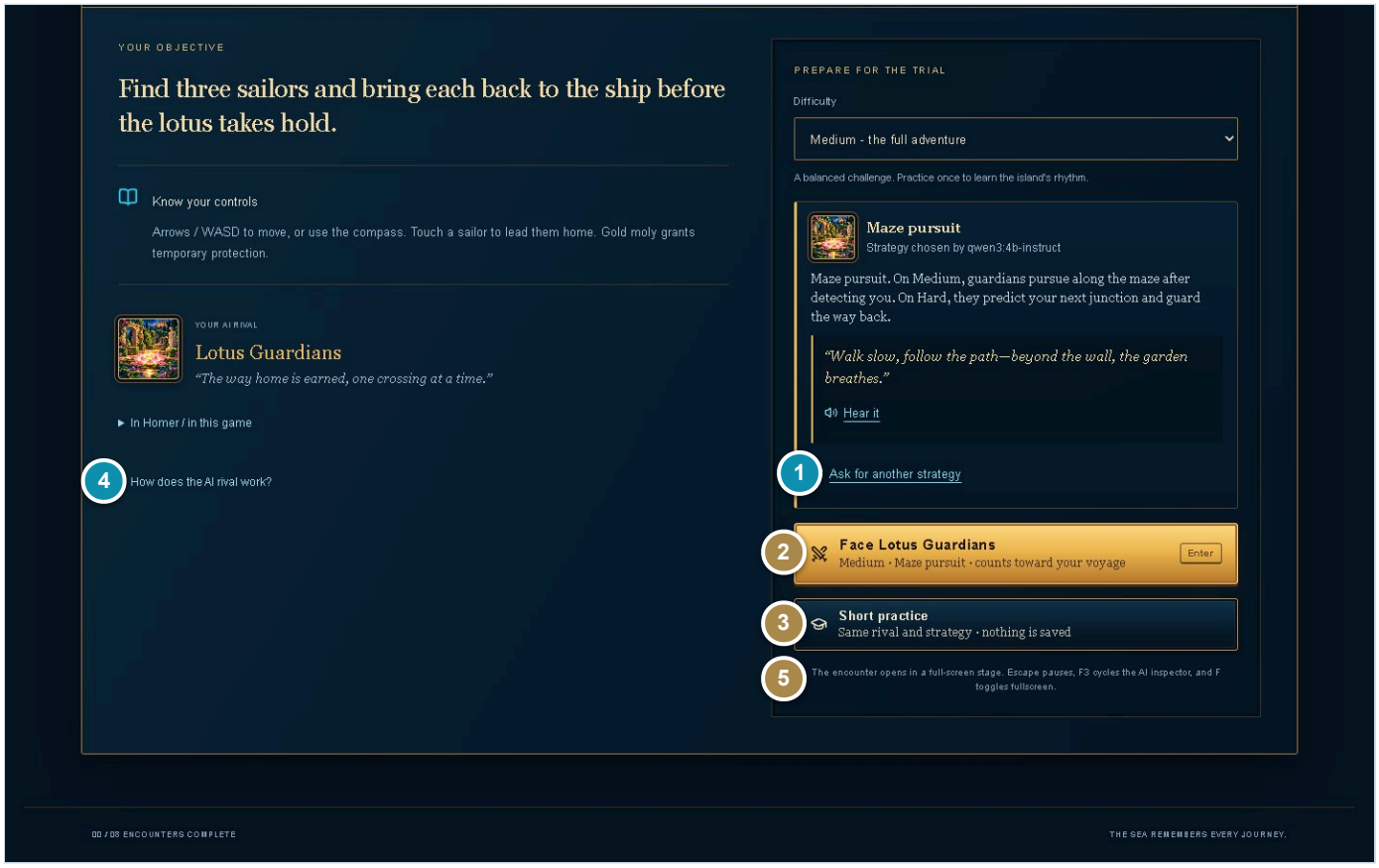



**AI state:** User control: the player asked for another strategy (Counter-gusts became Gentle breeze) and can undo. Mock AI service in mode success

1. The new strategy, and who chose it: "Strategy chosen by" the model.
2. Regenerate: Ask for another strategy.
3. Undo: back to the strategy the player had before. Pressing it again swaps back.
4. The start button repeats the chosen strategy, so the player knows what they are accepting.
5. Edit is not offered by design: the strategy is the opponent's move. Changing difficulty asks for a new one.

## D08 Briefing: controls and start

F3 Prepare the rival (A1 success) · Daniel, Rosa

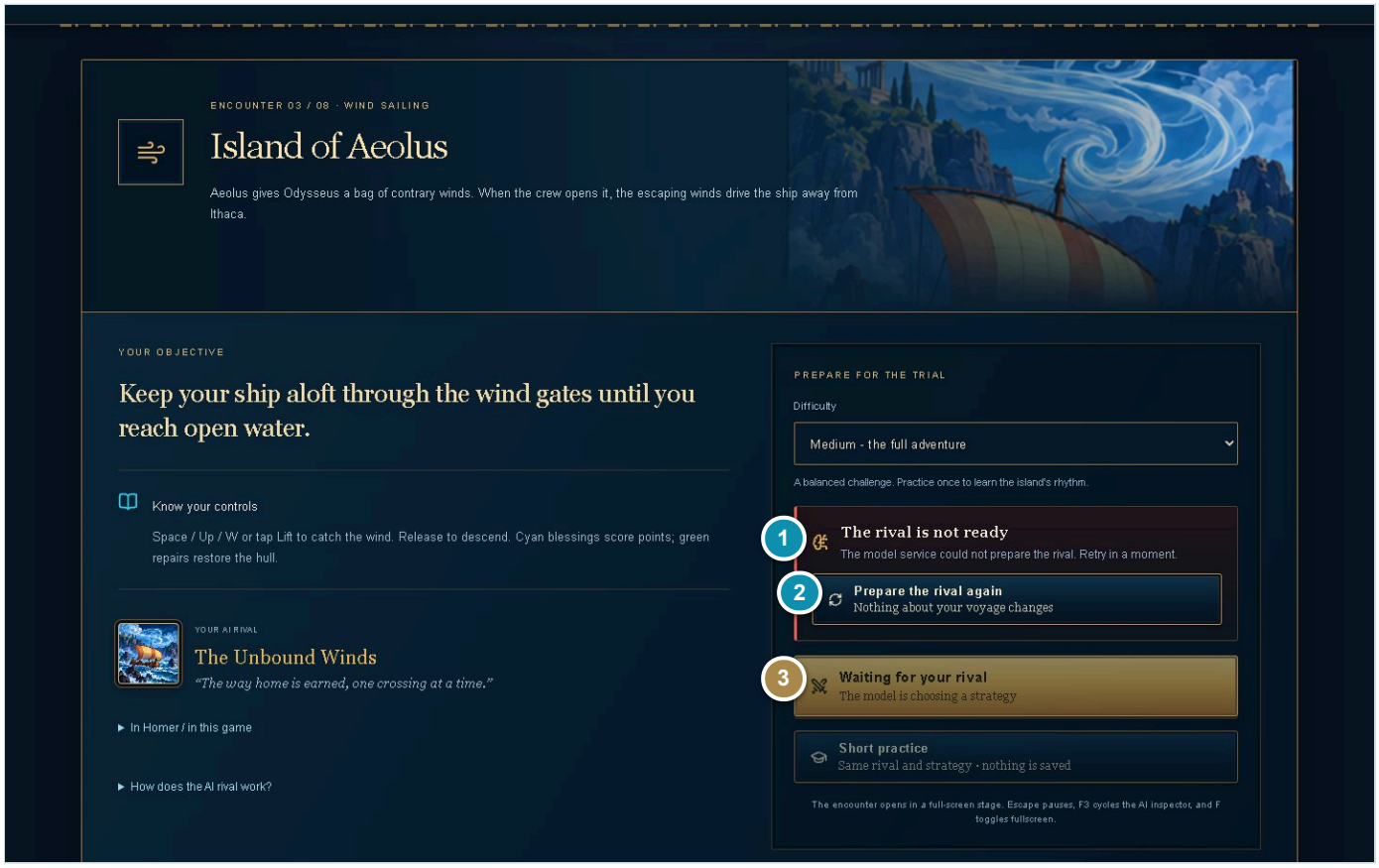



**AI state:** User control over the AI result: regenerate, start, or practise

1. Ask for another strategy: regenerate the AI result.
2. Start repeats difficulty and plan and says it counts toward the voyage. Enter is the shortcut.
3. Short practice: same rival and strategy, nothing saved.
4. "How does the AI rival work?" says what the model decides and what the game always owns (collisions, scoring, progress).
5. Explains Escape, F3 and F before the full-screen stage opens.

## D09 Briefing: AI error (state 5 of 5)

F4 Recover from a failed AI plan (A1 error) · Rosa, Daniel

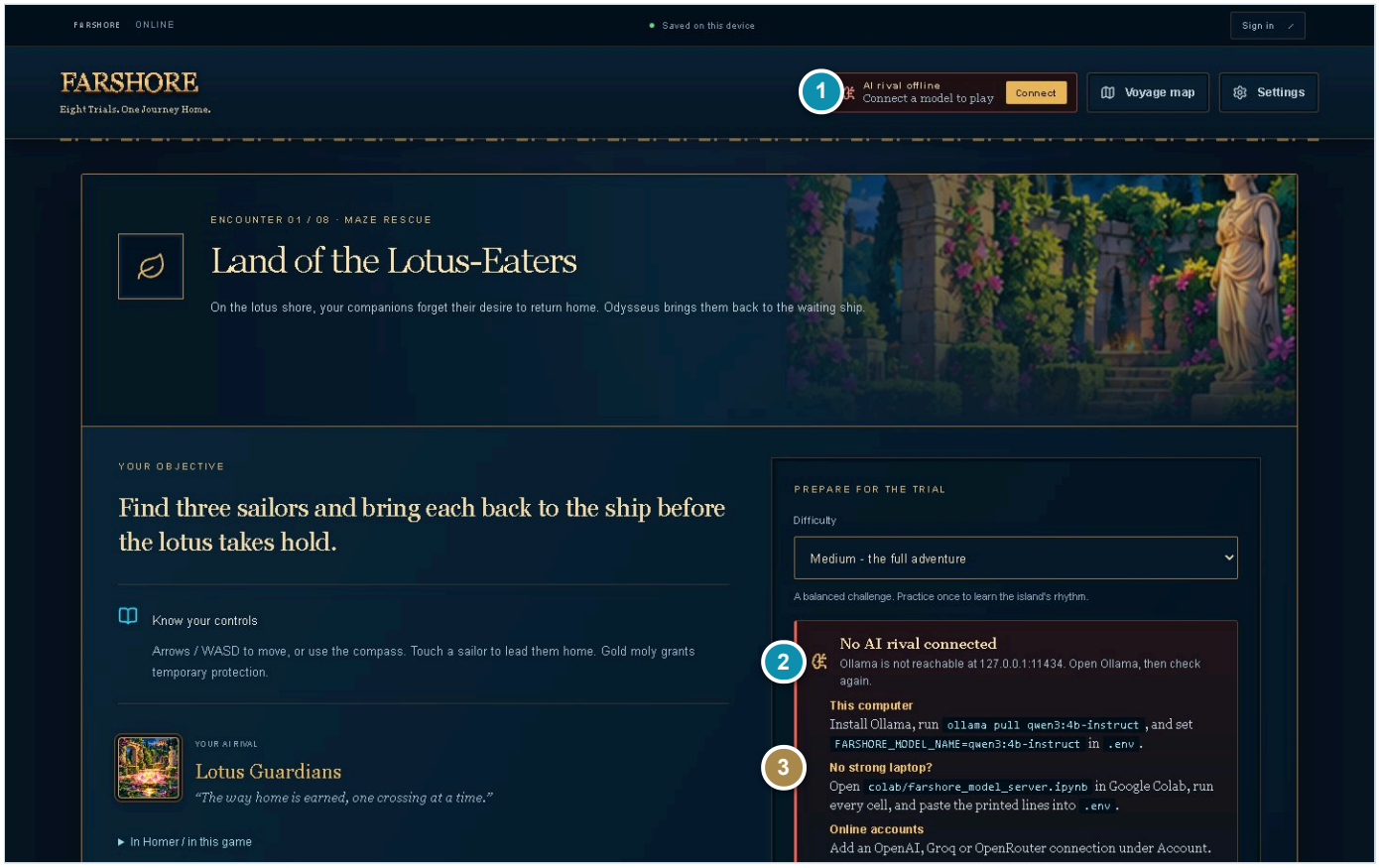



**AI state:** Error: the strategy request failed. Triggered with the mock AI service (mode error-once), which is how Task 4 of the test reaches it

1. A plain message with no error codes, in the red error style.
2. One clear recovery action, "Prepare the rival again", plus the reassurance "Nothing about your voyage changes".
3. Start stays disabled. Pilot: its label "The model is choosing a strategy" is wrong while nothing is being prepared.

## D10 Briefing: no AI rival connected (initial state without a model)

F3 Prepare the rival (A1 unavailable) · Daniel, Rosa



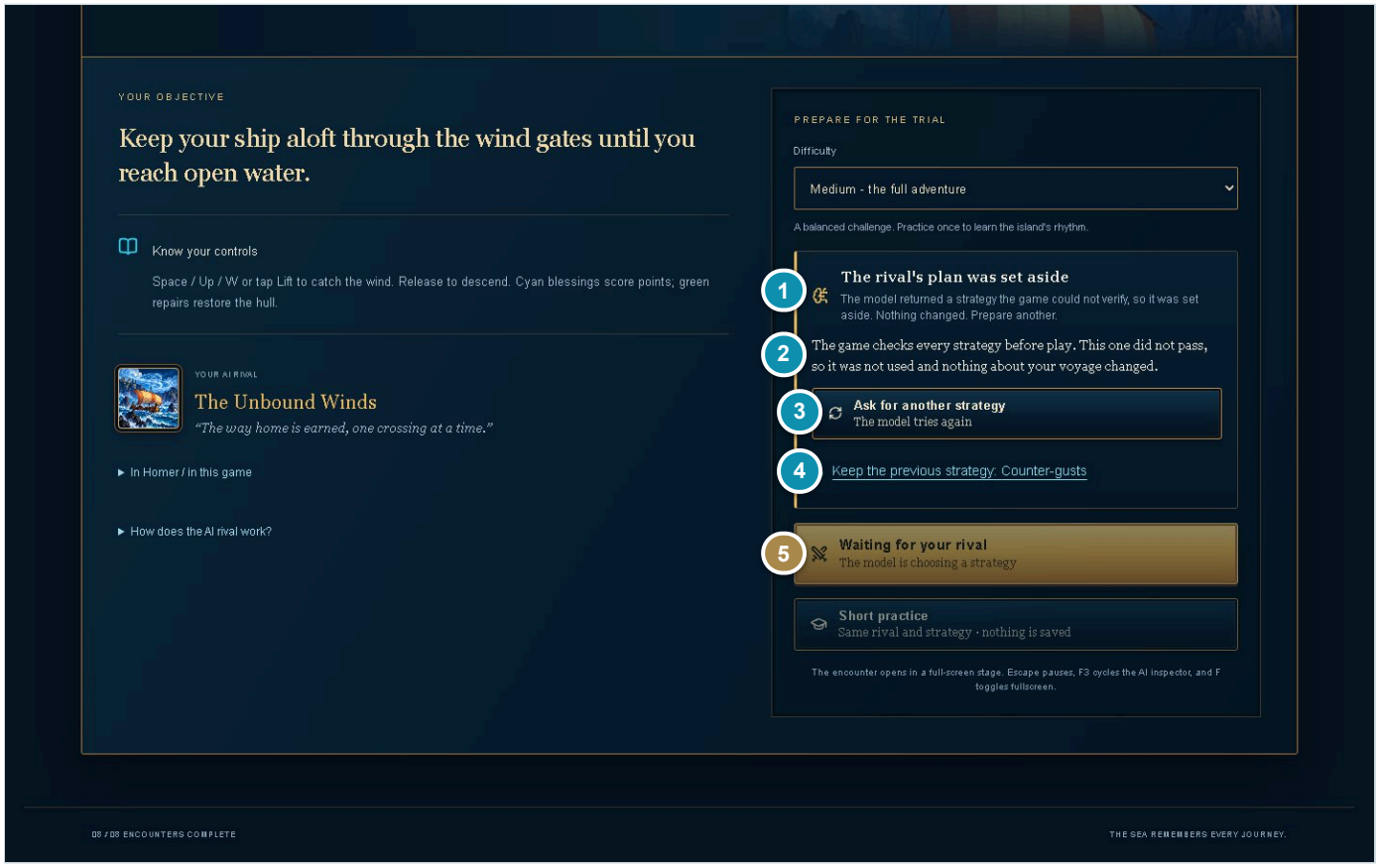

**AI state:** Model service unreachable: play is blocked until a model is connected

*Failure simulated for this capture (the status check reported the model as unreachable (the mock AI service's offline mode produces the same screen)); the interface is real.*

1. The header chip turns to "AI rival offline", with Connect.
2. Names the exact problem and the next step: open Ollama, then check again.
3. Three ways to connect: this computer, Google Colab, or an online account.
4. Check again sits below the fold. The game also re-checks every 15 seconds (T4).

## D22 Briefing: low-confidence result (state 4 of 5)

F4 Recover from a failed AI plan (A1 uncertain) · Rosa, Maya

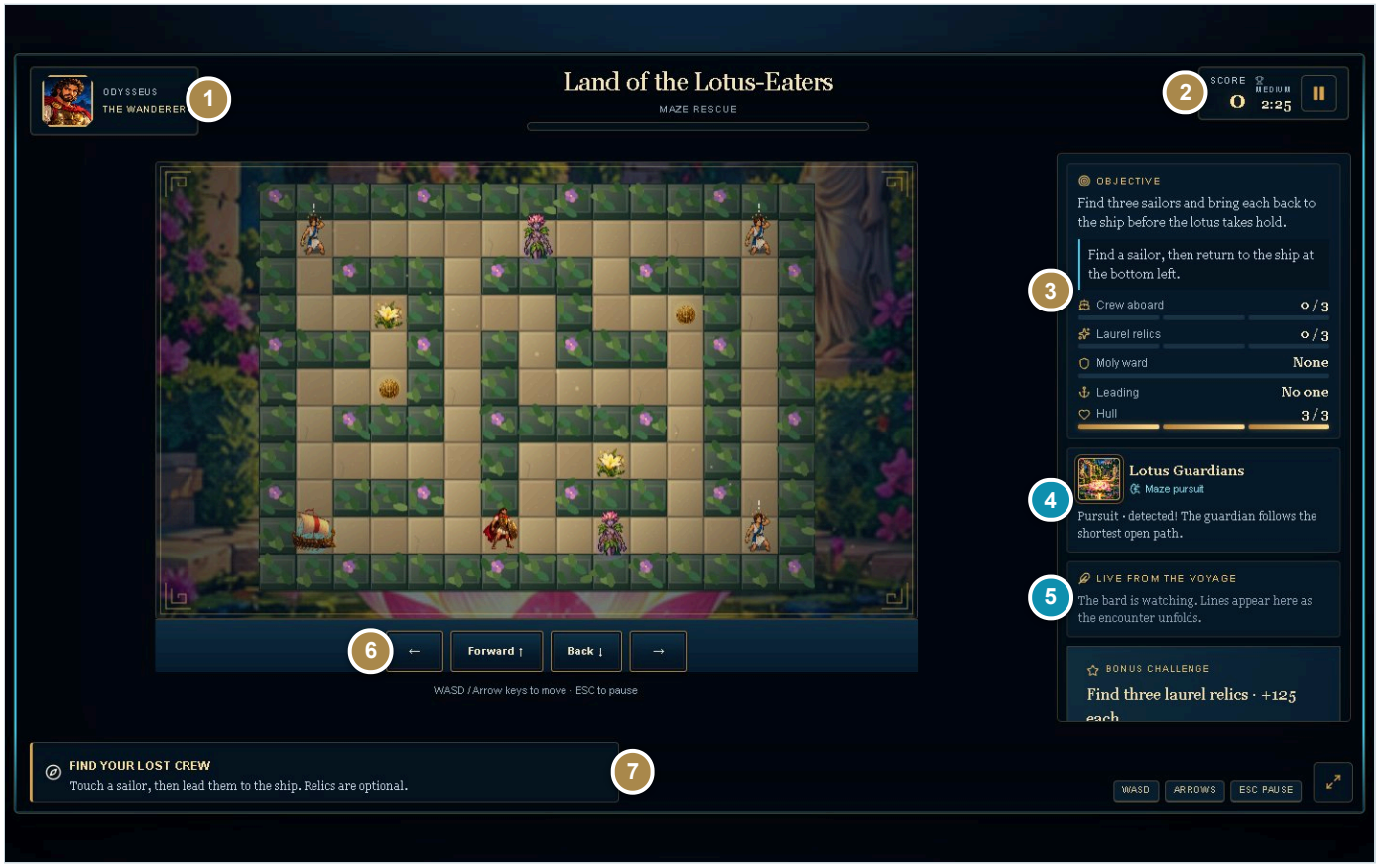



**AI state:** Low-confidence: the model returned a strategy that failed the game's checks, so it was set aside. Triggered with the mock AI service (mode uncertain-once) after the player asked for another strategy

1. Amber, not red: the AI produced an answer, but the game could not trust it.
2. Explains the check in plain words. Large language models give no reliable confidence number, so none is invented.
3. Ask for another strategy: the user regenerates the result. The rejected plan is never used.
4. Undo: Keep the previous strategy returns to the plan the player had before asking again.
5. Pilot issue 11: the disabled start button still says "The model is choosing a strategy" while nothing is being prepared.
6. New in Phase 3: before this, an unverifiable plan looked identical to a connection error.

## D11 Lotus-Eaters: playing

F4 Play · Maya, Daniel

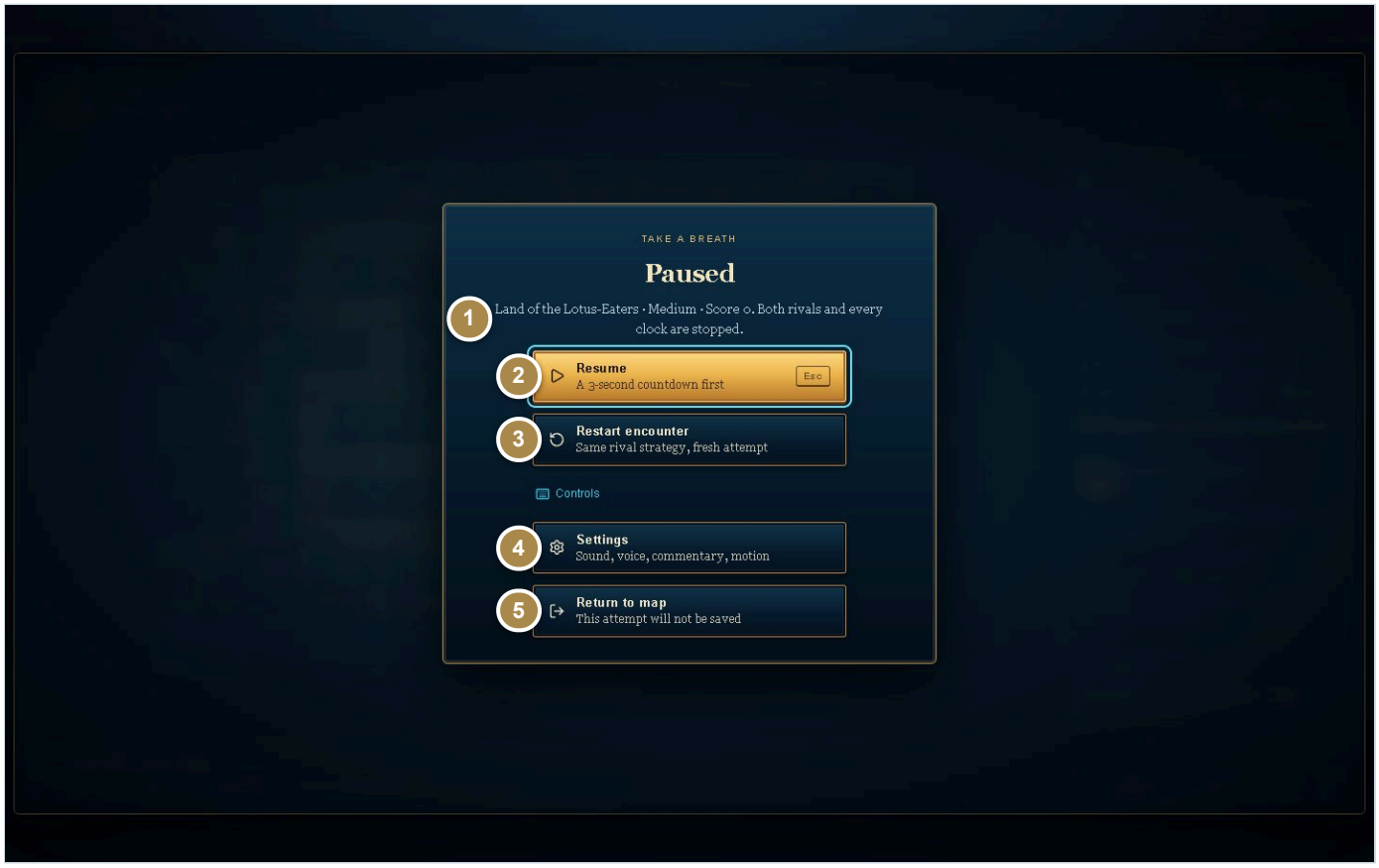



**AI state:** The chosen strategy runs locally every frame. Commentary lines are written by the model during play

1. Odysseus' portrait and lives, top left.
2. Score, difficulty, time and pause, top right.
3. Objective and progress counters stay in view.
4. Rival card: plan name and current behaviour. Pilot: the plan name and the live line can disagree.
5. Live from the voyage: the bard's and rival's lines appear here and are spoken.
6. On-screen direction buttons for mouse and touch players.
7. Tactical cue strip with the current instruction.

## D12 Pause

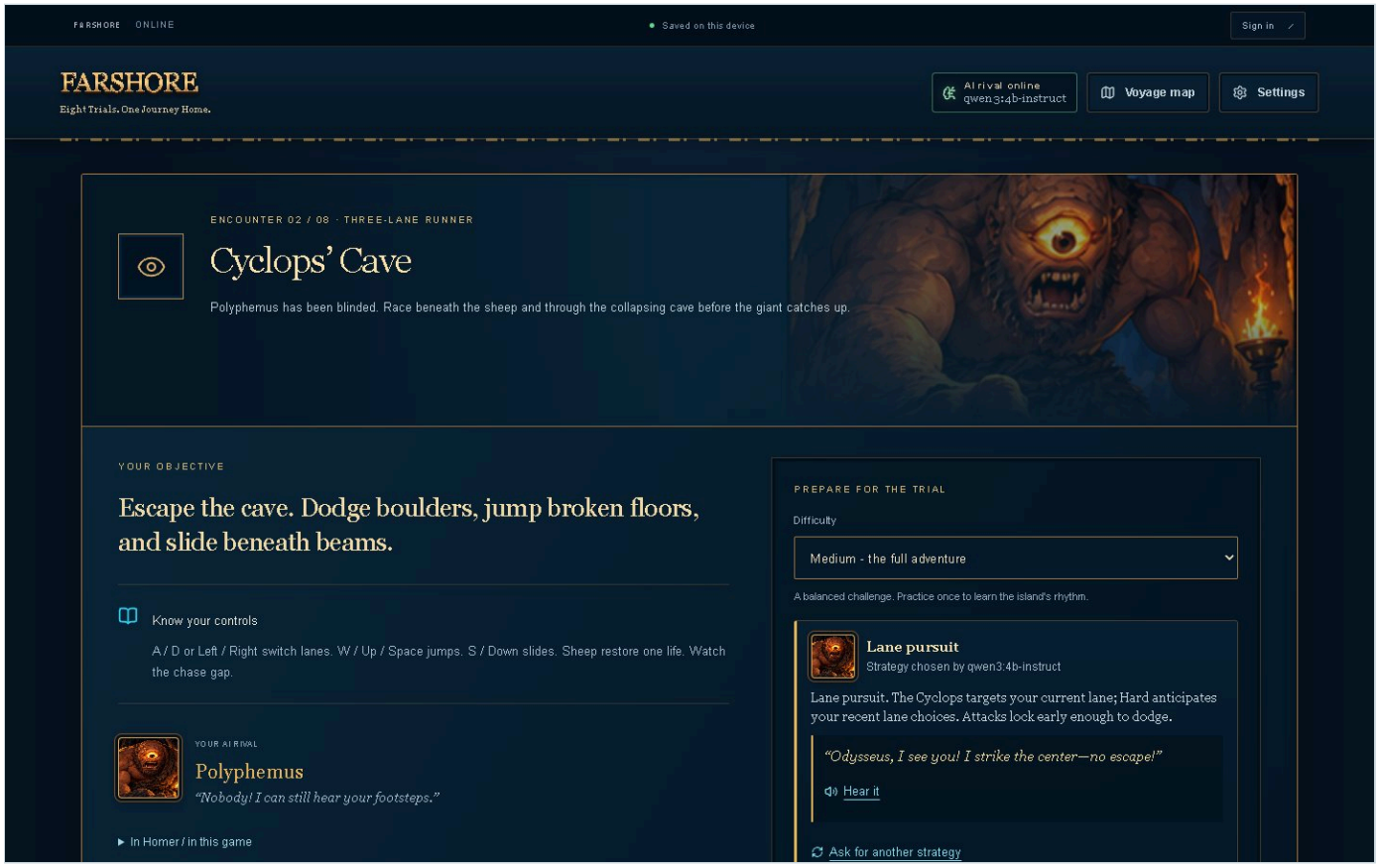
F4 Play · All players




1. States exactly what is paused: both rivals and every clock.
2. Resume has a 3-second countdown, so the player is not hit on return. Esc shortcut.
3. Restart keeps the same rival strategy.
4. Settings are reachable mid-play.
5. Return to map warns that the attempt will not be saved.

## D13 Cyclops' Cave briefing

F3 Prepare the rival (A1 success) · Maya

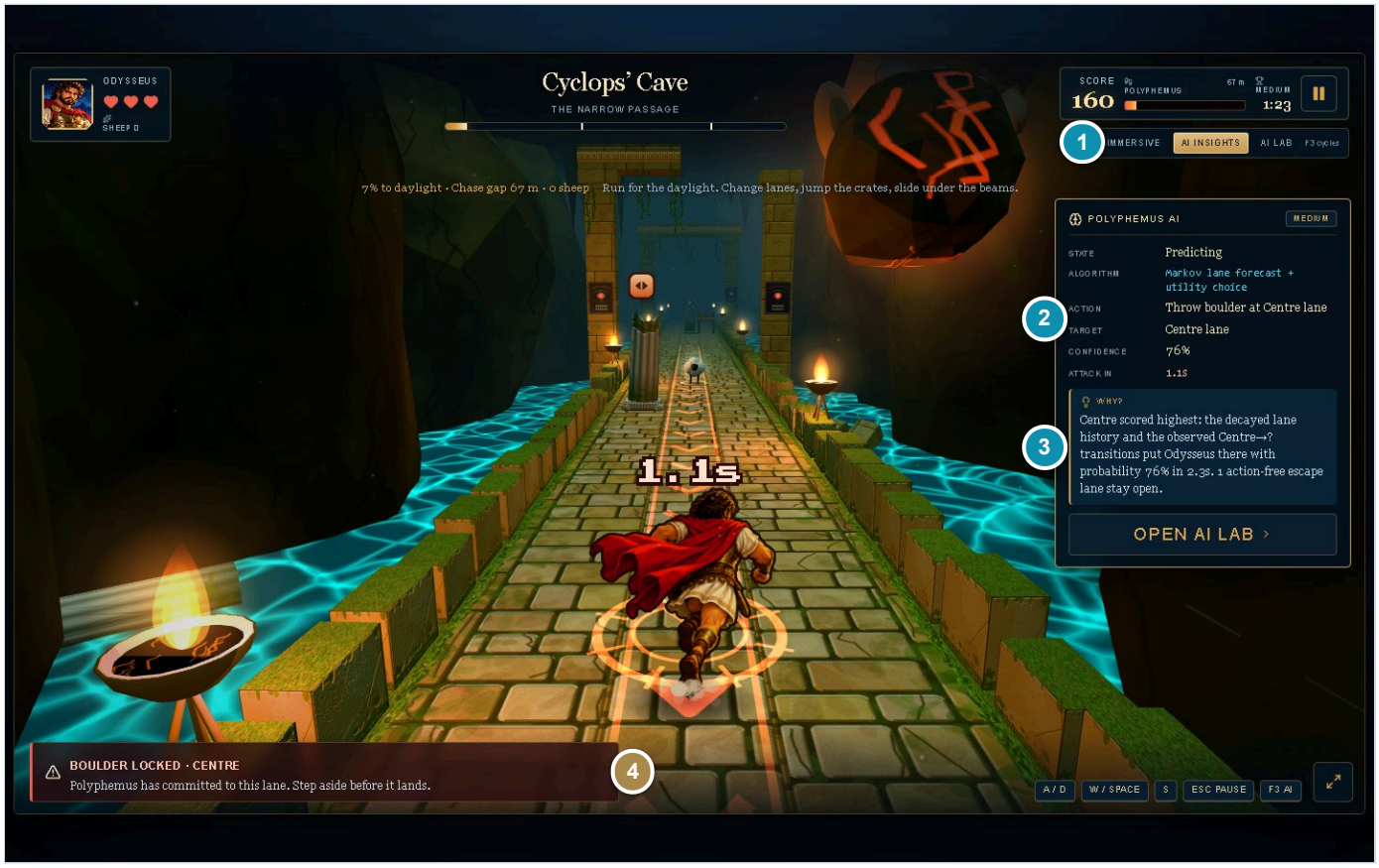



**AI state:** A1 success for Polyphemus

1. Same briefing layout for every trial, so learning carries over. The cave adds an Original cave tactics practice.

## D14 Cyclops' Cave: AI Insights

F5 Understand the rival · Maya, Rosa

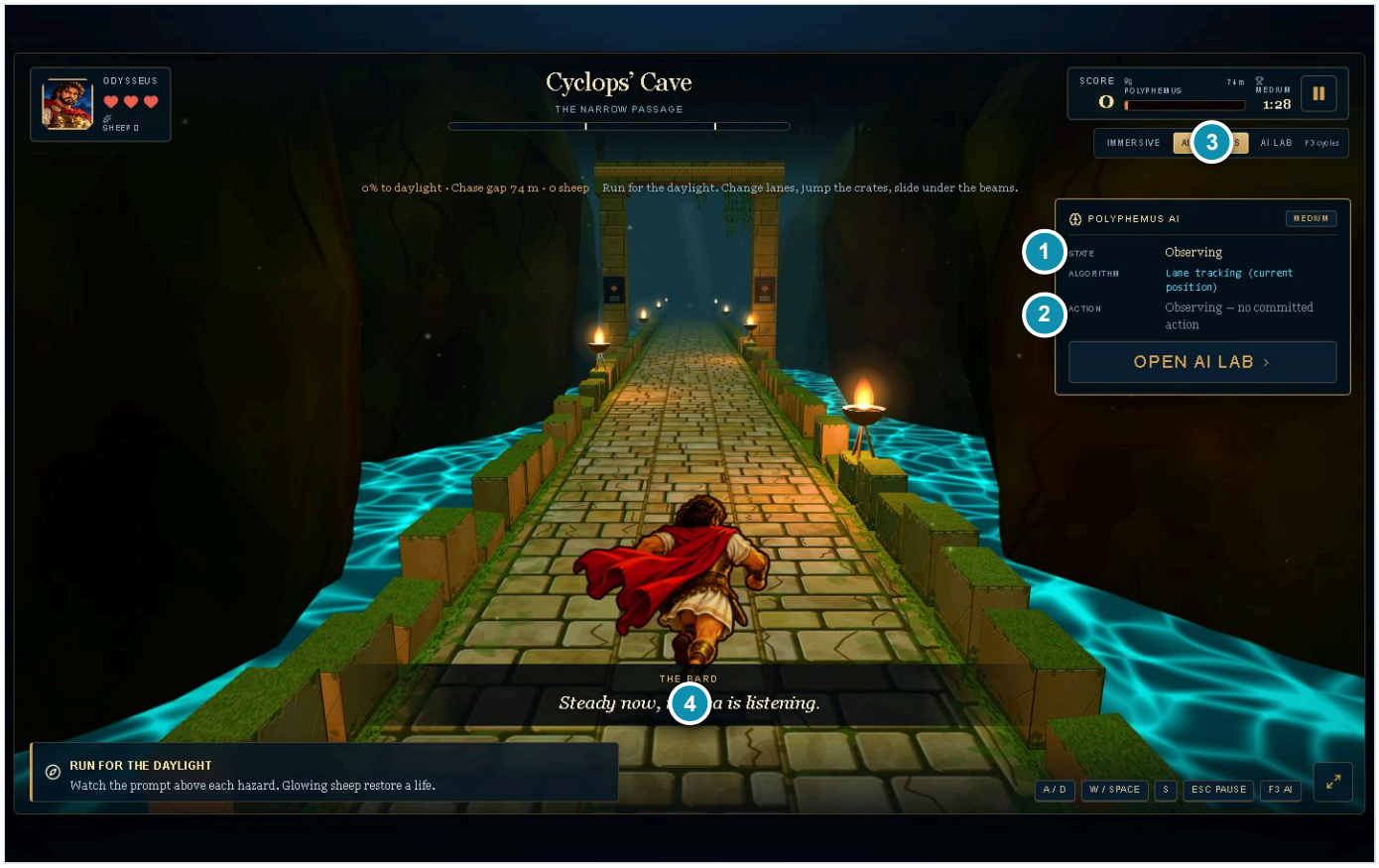



**AI state:** Explainable AI overlay: state, algorithm, chosen action and a grounded reason

1. Mode chips: Immersive, AI Insights, AI Lab. F3 cycles them.
2. State, algorithm, chosen action, target lane, forecast probability and time until the attack.
3. "Why?" gives the grounded reason, including the guarantee that an action-free escape lane stays open (T3).
4. The danger cue is red and in words: "Boulder locked, Centre. Step aside before it lands."
5. The 76% is a real Markov-model forecast of the player's lane, not an invented confidence. Cyan is used only for AI information.

## D23 Cyclops' Cave: AI Insights before the first decision (state 1 of 5: empty)

F5 Understand the rival · Maya, Rosa

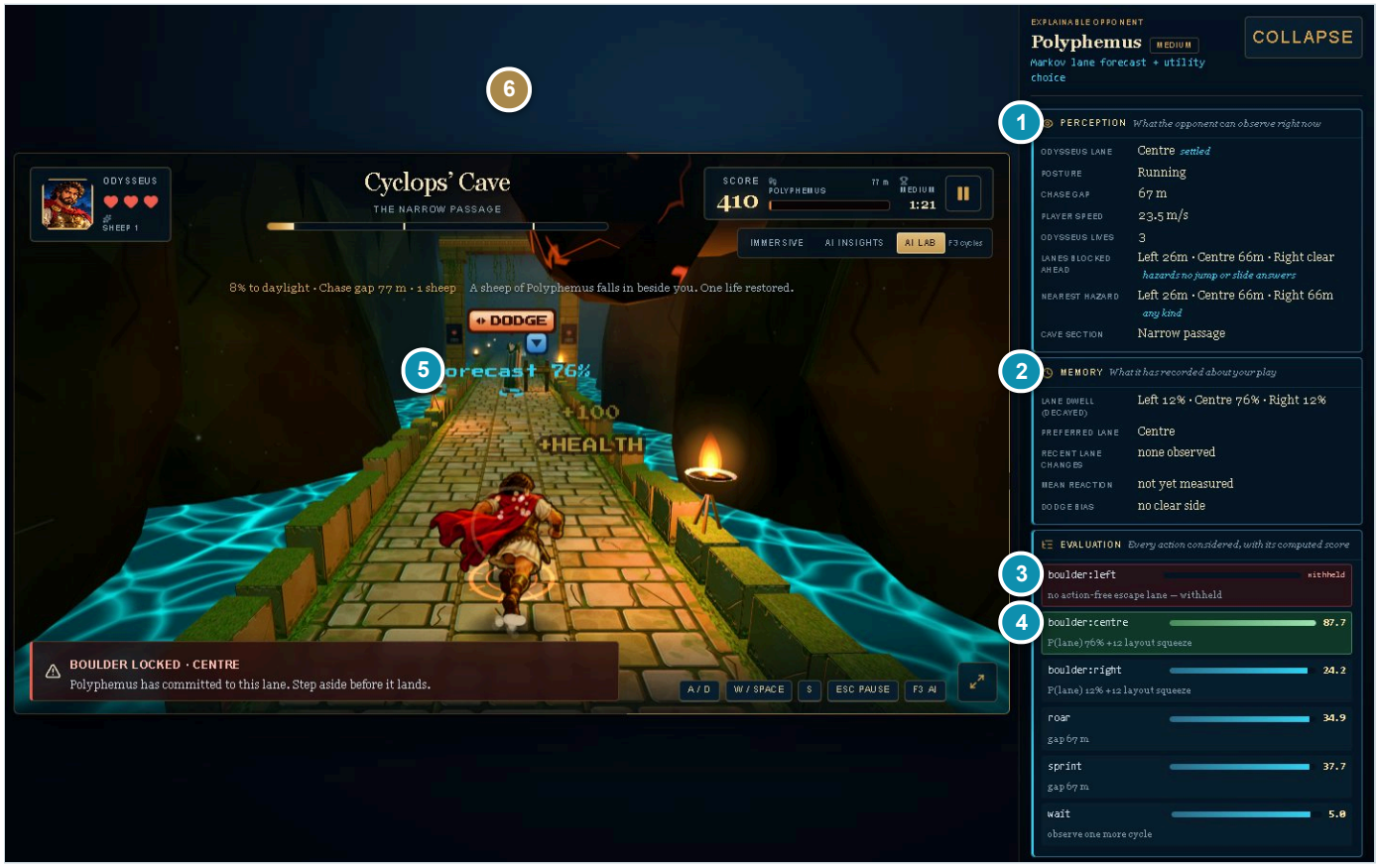



**AI state:** Empty state of the explainable AI panel: the rival is observing and has committed to nothing yet

1. State: Observing. The panel says what the rival is doing even before it acts.
2. "No committed action": an honest empty state, not a blank box or a made-up reason.
3. AI Insights is on by default in this encounter. F3 cycles Immersive, AI Insights and AI Lab.
4. Live bard commentary. In this capture the line comes from the mock AI service.

## D15 Cyclops' Cave: AI Lab panel

F5 Understand the rival · Rosa

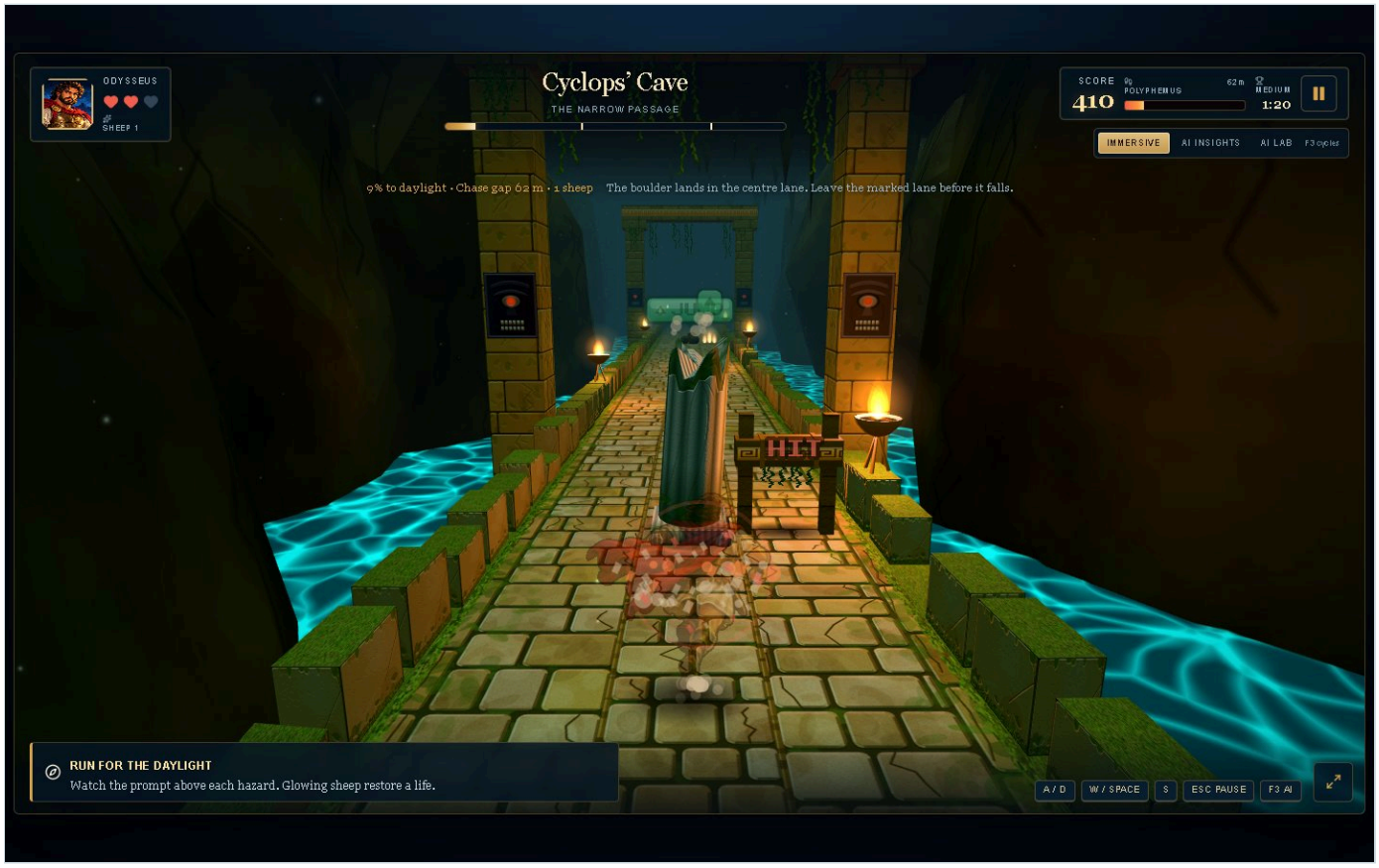



**AI state:** Full decision pipeline: perception, memory, evaluation, decision, execution, result

1. Perception: exactly what the opponent can observe right now.
2. Memory: what it has recorded about the player's habits (lane dwell, preferred lane).
3. A withheld option is shown dimmed, with its fairness reason (no action-free escape lane).
4. Every candidate action with its computed score. The chosen one is highlighted.
5. The forecast is also drawn in the scene over the targeted lane.
6. The play area shrinks to make room and returns to full width when the panel is collapsed.

## D16 Cyclops' Cave: Immersive

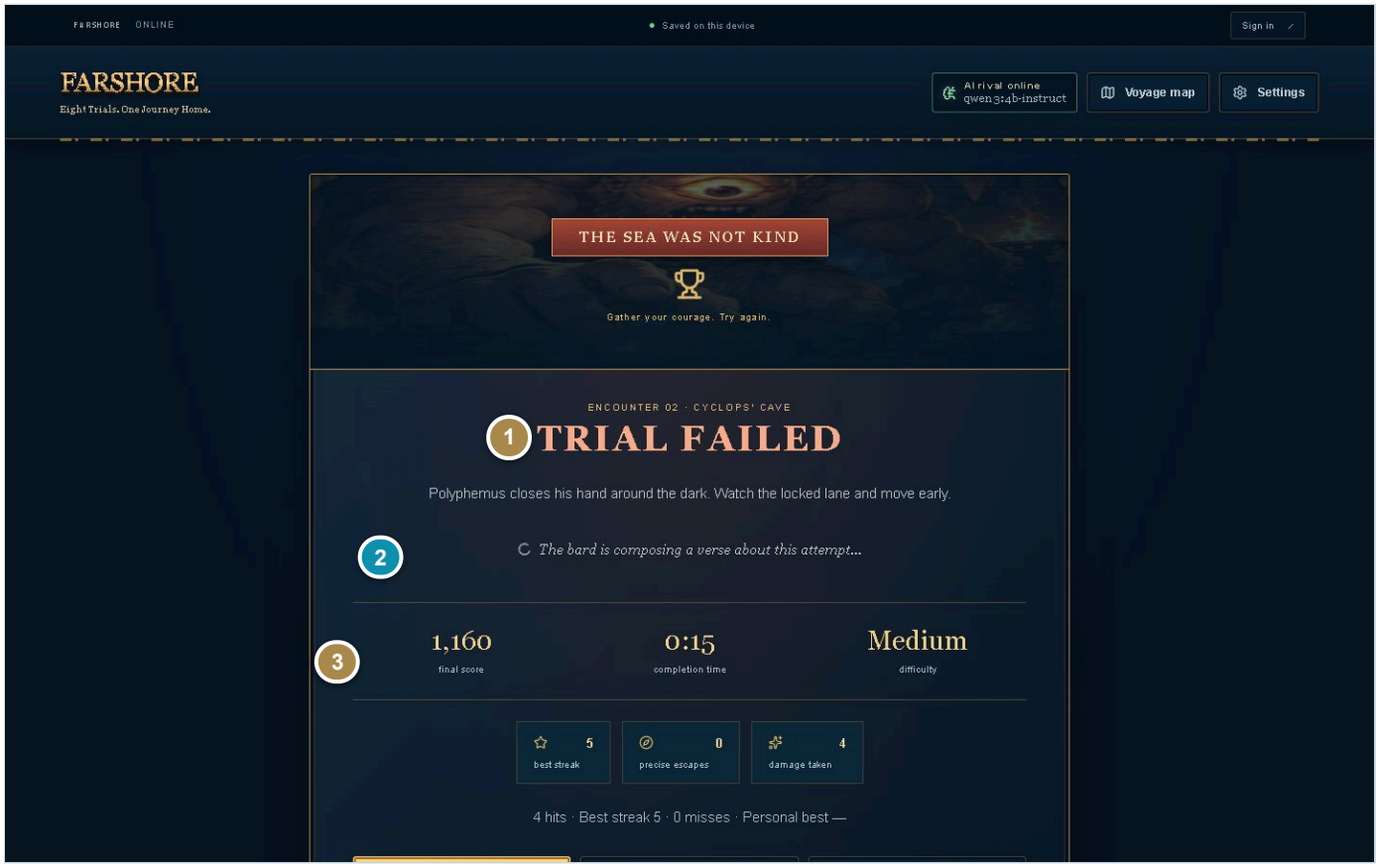
F4 Play · Maya




1 Immersive hides the inspector. Fair attack warnings (the locked-lane stripe and countdown) always stay, because they are gameplay information, not AI debugging.

## D17 Result: trial failed

F4 Play · Daniel, Maya

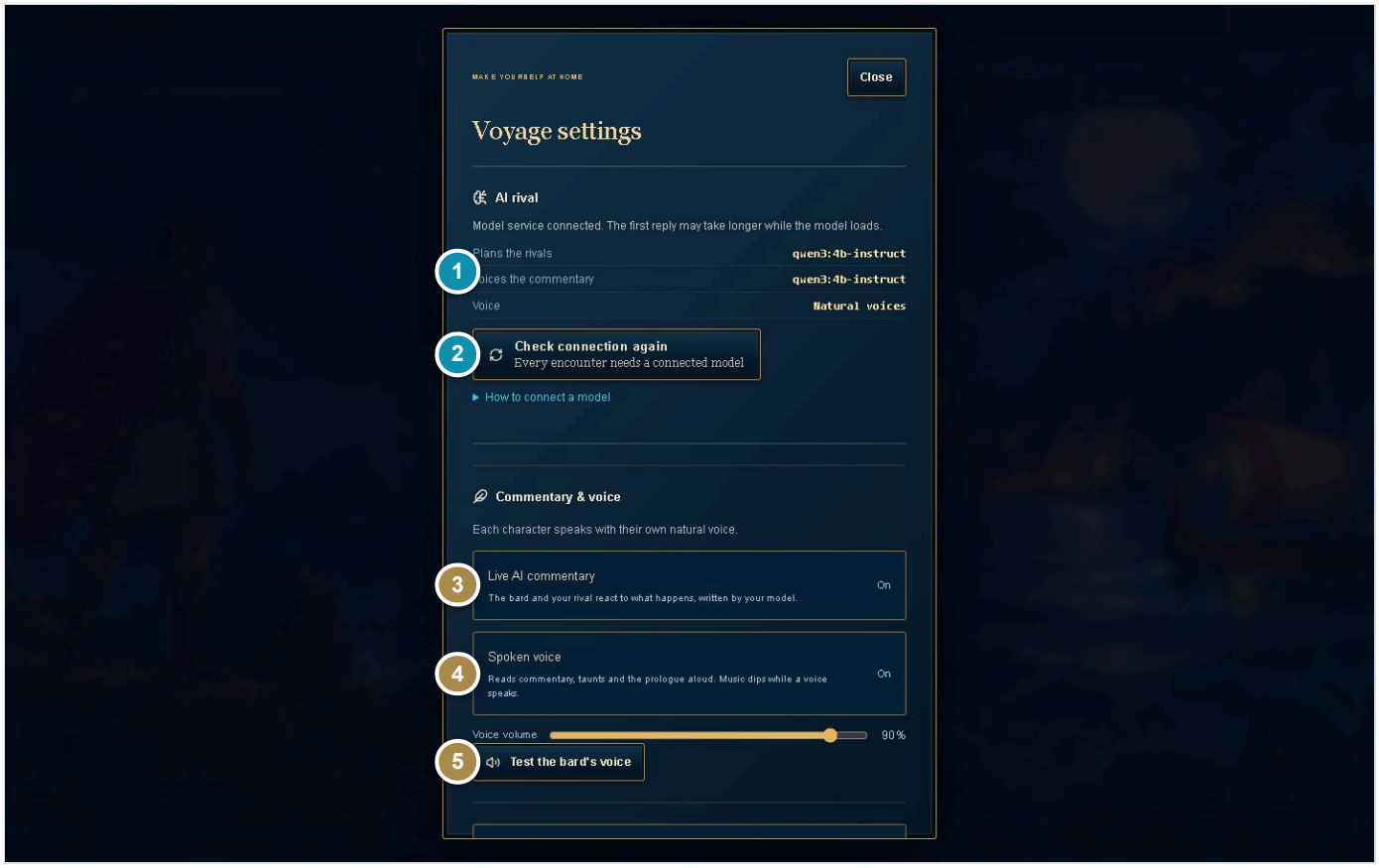



**AI state:** Model-written rival line reacting to the attempt

1. Clear outcome, with a specific tip: "Watch the locked lane and move early".
2. The rival reacts to this attempt in a line written by the model.
3. Score, time and difficulty.
4. Retry, Change difficulty, World map and an accessibility Assisted passage sit below.

## D18 Settings

F6 Settings · Daniel, Rosa

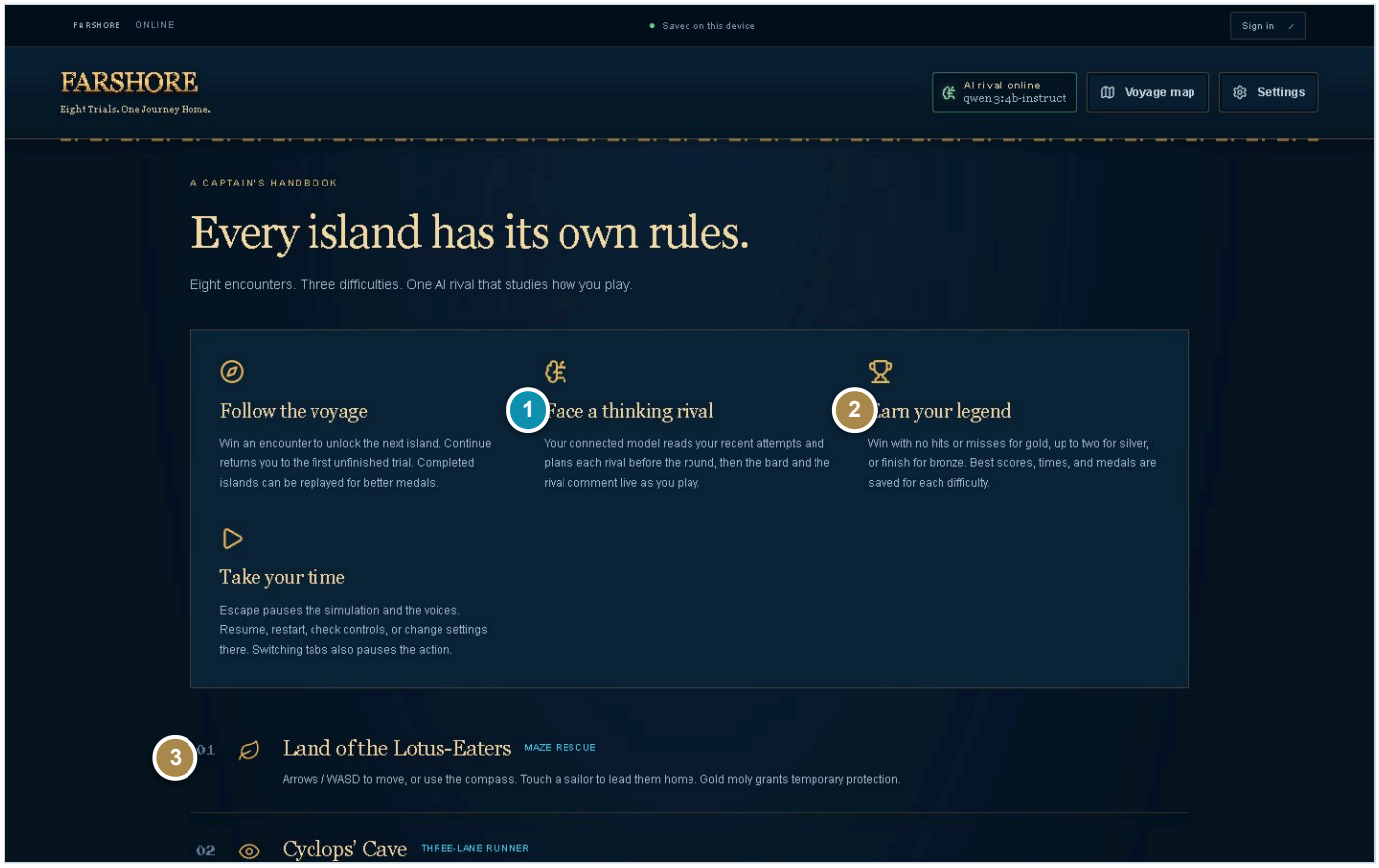



**AI state:** Which model plans the rivals and which voices the commentary

1. Names the models in use for planning, commentary and voice.
2. Check connection again, with the reminder that every encounter needs a model.
3. Live AI commentary on or off.
4. Spoken voice on or off, separately from commentary.
5. Voice volume and a voice test.

## D19 How to play

F2 Learn · Daniel

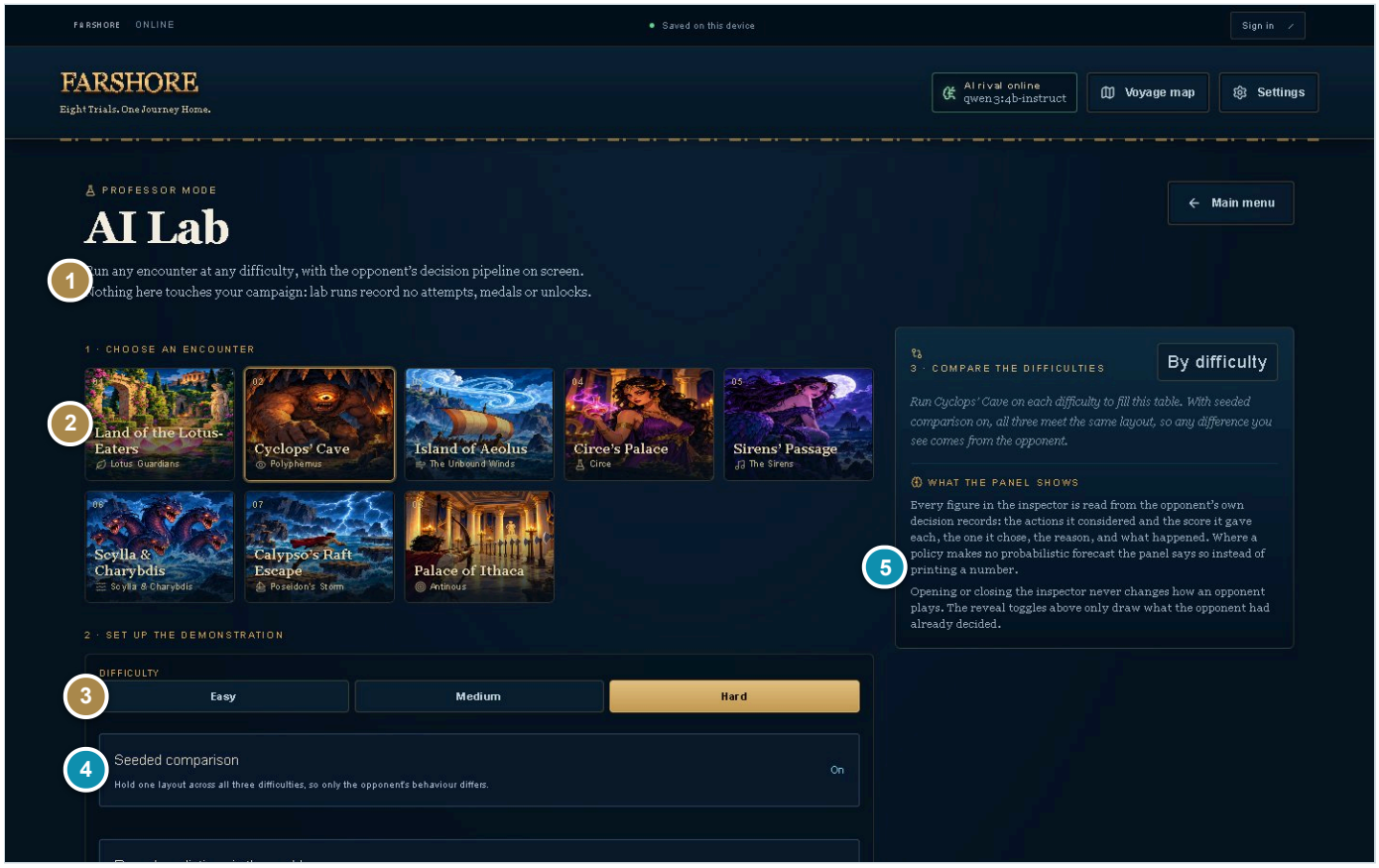



**AI state:** Explains the AI rival in one sentence

1. "Face a thinking rival": the model reads your attempts and plans the rival, then comments live.
2. The medal rules are stated exactly.
3. Controls for all eight trials on one page.

## D20 AI Lab (professor mode)

F5 Understand the rival · Rosa

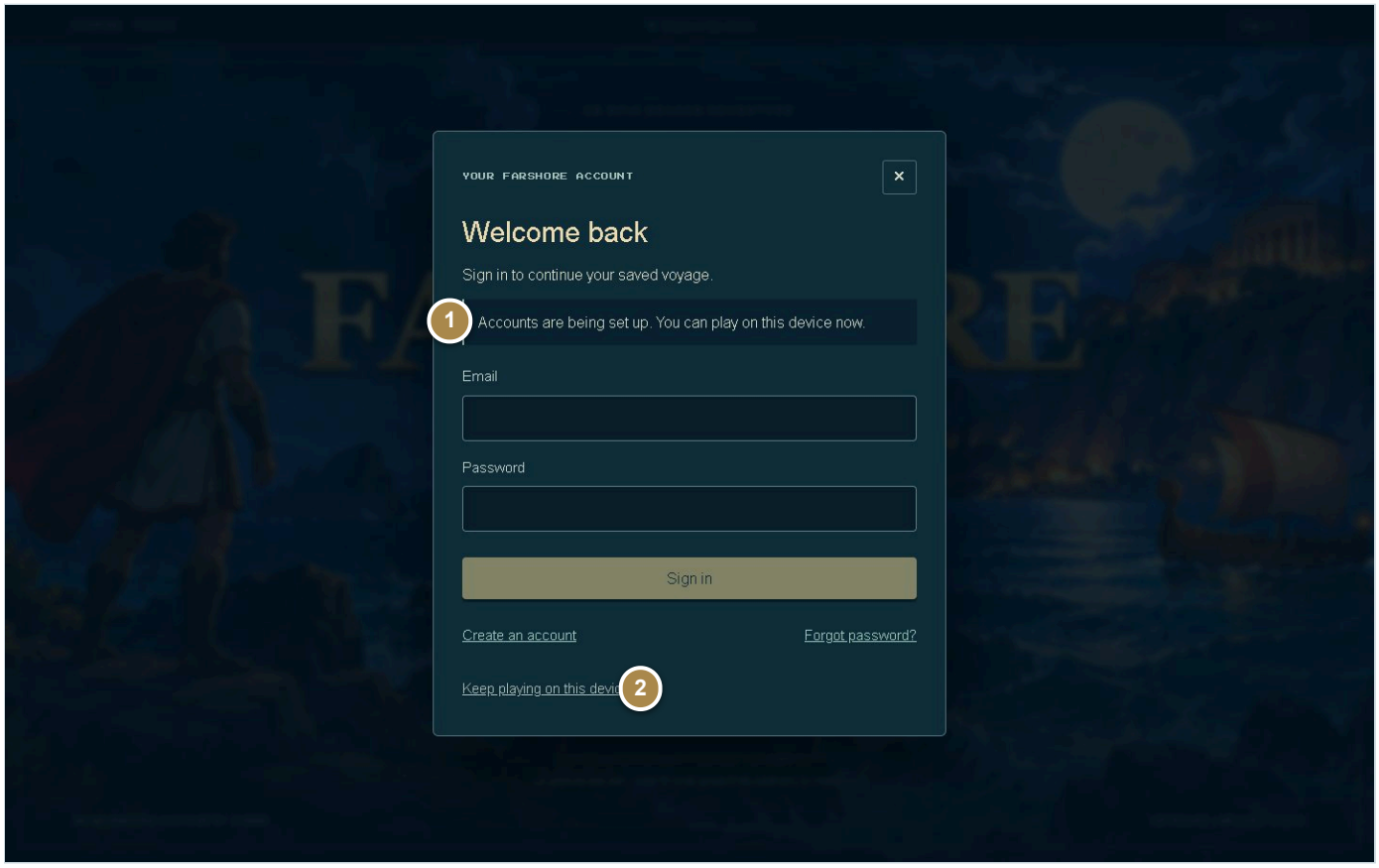



**AI state:** Compare the rival across difficulties with the same layout

1. States that lab runs never touch campaign progress.
2. Choose any of the eight encounters.
3. Choose a difficulty.
4. Seeded comparison keeps the layout fixed, so any difference comes from the opponent.
5. Says the inspector never changes how the opponent plays.

## D21 Sign in

F7 Account · Maya

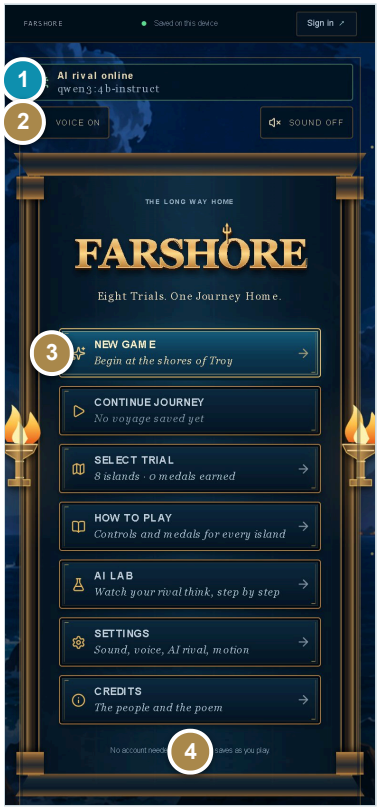



1. When accounts are not configured, says so and offers play on this device.
2. "Keep playing on this device" is always available.
3. Email/password and Google sign-in sync saves through MongoDB when the server is configured.

## 1.4 Phone screens (390 × 844)

### M01 Title menu (phone)

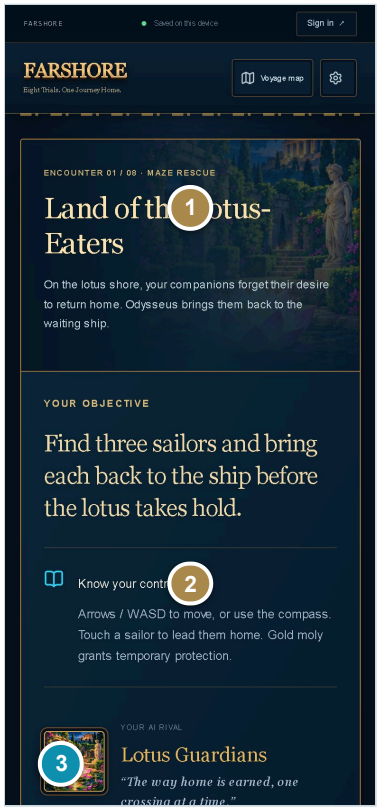
F1 Start · All players




1. The AI rival status gets its own full-width row at the top.
2. Voice and sound toggles follow directly below.
3. The menu stacks into one column, and every entry keeps its one-line explanation.
4. "No account needed. Your voyage saves as you play."

## M02 Briefing (phone)

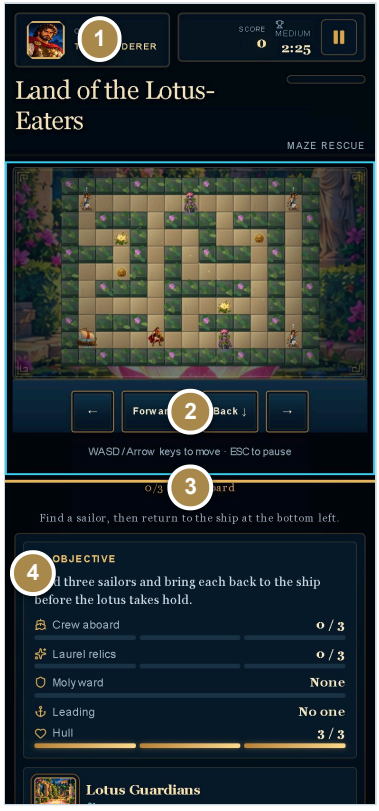
F3 Prepare the rival (A1 success) · Daniel




1. Story and objective come first, in large type.
2. Controls are explained in words, including touch.
3. Pilot: on a phone the rival's strategy panel and the start button are several screens further down.

## M03 Lotus-Eaters (phone)

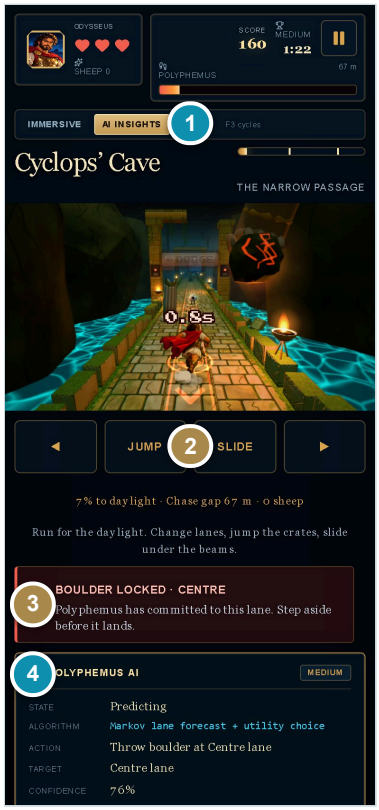
F4 Play · Maya, Daniel




1. Portrait, then score, time and pause: the same HUD order as desktop.
2. Large on-screen direction buttons for touch.
3. Progress (crew aboard) and the current instruction sit directly under the maze.
4. Objective and counters follow below.

## M04 Cyclops' Cave with AI Insights (phone)

F5 Understand the rival · Maya, Rosa




**AI state:** Insights panel below the play area

1. Mode chips: Immersive, AI Insights, AI Lab.
2. Touch controls: lane left and right, Jump, Slide.
3. The locked-lane warning is in words as well as colour.
4. Pilot: the Insights panel sits below the play area on a phone, so it cannot be read while dodging.

# 2 Test script and consent

Cyclops-only adaptation of the original Farshore protocol. The four goals below match the supplied participant JSON files. P1's setup evolved during the session; that deviation is retained in the results. Use the documented script consistently for the Phase 5 comparison.

## Session fields and setup

Use IDs P1-P5. Record the date, exact build or commit, device, difficulty, moderator, consent and recording permission. The uploaded records do not supply session dates or exact versions. Recording permission is reported Yes for P2-P5 and unspecified for P1; that does not establish that recordings were made.

Launch the local game and AI service using the project README. Use a fresh private browser session for each participant and warm the strategy service before timing. Keep the same game build and script across participants. Time navigation and explanation tasks separately from the duration of a gameplay attempt.

## Read aloud before tasks

You are helping test Farshore for a university assignment. This takes about 20 minutes. Your answers will be anonymous. You may skip questions or stop anytime. No recording is needed. Please avoid sharing personal information. Do you agree to participate?

Record consent before proceeding. If audio or video is requested, obtain separate permission and explain access and retention. Do not include names or recordings in AI prompts.

## Warm-up questions

1. How often do you play video games, and what kind?
2. Against a computer opponent, how do you work out what it is planning?
3. Have you used AI tools such as ChatGPT, or games with AI-driven characters before?

## Moderator rules

Ask participants to think aloud. Stay quiet, avoid hints and do not point out controls. Use neutral prompts such as What are you thinking? Do not explain the AI or defend the design during tasks. Record any assistance. Stop if a participant wishes to stop or is distressed.

# 2 Task goals and measurements

Start timing when the participant begins the stated task and stop when they finish or give up. Record Yes, Partly, No or Gave up, time in seconds, errors or wrong turns, assistance and observations. After T1-T3 ask ease from 1 very difficult to 5 very easy, plus why.

### T1 Open the Cyclops briefing

Task: You have just opened Farshore. Start a new journey and get ready for Cyclops' Cave.
Success criterion: the Cyclops trial briefing is reached. Do not require winning the encounter.

### T2 Locate the opponent plan

Task: Before facing the Cyclops, find what the opponent plans to do and who or what decided that plan.
Success criterion: locate the plan, describe it and identify its source. Record the participant's actual attribution; do not supply the answer.

### T3 Explain a Cyclops decision

Task: Use what the game displays to explain why the Cyclops made one specific move.
Success criterion: describe a particular action and the displayed reason. Record whether the panel was noticed without prompting and whether any assistance was needed. Preserve the explanation as the participant's interpretation.

### T4 Recover from preparation failure

Task: Get ready to play Cyclops' Cave again.
Moderator setup, outside the timed task: trigger an AI preparation failure using the mock service in a second tab of the same browser session. Do not warn the participant. In P1, requesting another strategy was needed before the briefing displayed the failure.

Time recovery from handoff after the error is visible until a ready-to-play state is restored, or the participant gives up. Record noticing, next action and independent recovery. Accept a successful retry or valid restoration of the previous plan. Reset mock mode after the task. T4 ease was collected for P2-P5 only and is supplementary.

### Separate gameplay observation

Record the runner attempt duration, win or loss, health lost, interface errors and gameplay ease separately. A death from rocks does not automatically mean that locating or explaining the AI information failed.

# 2 Questionnaires and session close

After all tasks, administer the instructor's ten SUS statements unchanged. Use 1 strongly disagree to 5 strongly agree. Do not explain or reword statements; if asked, say whatever it means to you.

| # | Statement |
|---|---|
| 1 | I think that I would like to use this app frequently. |
| 2 | I found the app unnecessarily complex. |
| 3 | I thought the app was easy to use. |
| 4 | I think that I would need the support of a technical person to be able to use this app. |
| 5 | I found the various functions in this app were well integrated. |
| 6 | I thought there was too much inconsistency in this app. |
| 7 | I would imagine that most people would learn to use this app very quickly. |
| 8 | I found the app very awkward to use. |
| 9 | I felt very confident using the app. |
| 10 | I needed to learn a lot of things before I could get going with this app. |

Scoring: odd items contribute answer minus 1; even items contribute 5 minus answer. Multiply the sum by 2.5. SUS is a score out of 100, not a percentage. The instructor template gives about 68 as the comparison average.

## AI trust and perception

Ask trust from 1 not at all to 5 completely, plus why. Ask how results were judged, what they thought happened during trouble, what they wanted to change and whether they would use the app instead of their current approach. P1 supplied the trust rating and reason; the other perception fields are missing.

## Debrief and export

Ask what was most frustrating, what worked best and what was expected but missing. Thank the participant. Preserve the original wording separately from summaries. Export the JSON, save it to the study data folder, and transfer measured values to the workbook without estimating missing entries.

# 3 Study method and participants

Five participants each completed a Cyclops' Cave session on the local Farshore build, following the script in Section 2. The moderator recorded each session as a JSON file with task results, all ten SUS answers, the trust rating and notes. All five records report consent.

| ID | Reported background | Measurement coverage |
|---|---|---|
| P1 | Daily gamer; GTA and other games | T1/T4 recorded; T2/T3 outcomes and times missing |
| P2 | Regular gamer; frequency conflicts between structured and raw fields | Four task logs and questionnaires |
| P3 | Casual mobile gamer; limited AI experience | Four task logs and questionnaires |
| P4 | Daily expert gamer; experienced with AI tools | Four task logs and questionnaires |
| P5 | Regular adventure and racing-game player | Four task logs and questionnaires |

## Data corrections by the moderator (November 5, 2026)

- **Confirmed as recorded for P2-P5:** task times, error counts, ease ratings, the ten SUS answers, trust ratings, gameplay times and results.
- **Not verbatim for P2-P5:** the free-text fields (warm-up answers, reasons for ratings, decision explanations, trust and debrief answers). An AI tool reworded them while the records were typed up. They are used only as paraphrased notes, and the backgrounds above come from them.
- **T4 recovery action:** the P2-P5 records name a "choose another strategy" button, which does not exist. The moderator confirmed that P2-P5 pressed *Keep the previous strategy*. P1's button is unconfirmed.
- The original JSON files are unchanged. The corrections are listed in `Raw-data/Data-corrections.md` and applied in the workbook.

## Scope

The game has eight trials; participants tested the Cyclops runner, its briefing, the opponent information and failure recovery. Findings apply to that scope. Screens for other trials remain design documentation and were not tested by this group.

## Analysis

Each metric uses its available-case denominator. Missing values are excluded, not converted to zero or failure. Reported task completion is kept separate from independent completion when assistance is documented. SUS is recomputed from all ten answers; all five recorded scores match. Trust uses the recorded 1-5 rating.

## Source integrity

The original JSON files are included unchanged. The workbook keeps field paths and the recorded wording in RawNotes, session metadata in SessionNotes, and gameplay separately from task time. Session dates and exact build versions were not recorded.

# 3 Task results

| Task | Reported Yes | Missing outcomes | Median seconds (n) | Mean ease (n) |
|---|---|---|---|---|
| T1 | 5/5 | 0 | 4 (5) | 4.2 (5) |
| T2 | 4/4 | 1 | 30 (4) | 3.2 (4) |
| T3 | 4/4 | 1 | 21 (4) | 2.8 (5) |
| T4 | 5/5 | 0 | 8 (5) | 3.8 (4) |

All 18 recorded outcomes are Yes, while two of the 20 planned task outcomes are missing. This is 100% reported completion among recorded outcomes, with 90% outcome coverage. It is not evidence that every task was completed independently: P3's T3 notes explicitly describe prompting.

## Entry and strategy understanding

T1 was completed in a median 4 seconds, with mean ease 4.2/5. T2 took a median 30 seconds across four timings. T3 took 21 seconds across four timings, while the mean panel-ease rating was 2.8/5 across all five participants. Locating and interpreting opponent information were less straightforward than starting.

## AI preparation recovery

All five records report recovery without help; the median reported time is 8 seconds, ranging from 4 to 14. Four noticing fields are supplied, all Yes; P1's noticing field is missing. P1's exact stopwatch boundaries were not separately confirmed. Supplementary T4 ease is 3.8/5 across four ratings.

| Participant | Runner seconds | Game result | Gameplay ease |
|---|---|---|---|
| P1 | 40 | Lost | 2 |
| P2 | 45 | Lost | 3 |
| P3 | 22 | Lost | 1 |
| P4 | 55 | Won | 4 |
| P5 | 38 | Lost | 3 |

One participant won and four lost. Median runner duration was 40 seconds. These gameplay outcomes and durations are separate from the task success and explanation times above. Difficulty and device equivalence were not established across records.

# 3 SUS and AI trust

| Participant | SUS / 100 | AI trust / 5 |
|---|---|---|
| P1 | 50.0 | 2 |
| P2 | 72.5 | 3 |
| P3 | 30.0 | 2 |
| P4 | 87.5 | 5 |
| P5 | 57.5 | 3 |

Mean SUS is 59.5, median 57.5, with sample standard deviation 21.9 and a range of 30.0 to 87.5. The mean is below the instructor template's 68 comparison value. The spread matters: the casual participant struggled, while the expert participant rated the interface highly.

Mean and median AI trust are both 3.0/5. One of five participants rated trust at 4 or 5. P1 said it "felt like it was guessing" (verbatim). The paraphrased notes for P2 and P3 describe the forecast as a guess after quick direction changes and as hard to connect with what the boss did; P4's note reports close correspondence with visible cues. These are reported perceptions, not an accuracy evaluation of the AI model.

The results favor a compact default explanation with optional deeper detail. They do not justify removing the detailed view valued by the expert participant. The fix in Section 4 keeps the full AI Lab and makes only the in-play panel compact.

# 3 Findings and severity

Severity is assigned during analysis: 0 no problem, 1 cosmetic, 2 minor, 3 major, 4 catastrophic. Counts refer to participant records containing the defined issue, not to every possible occurrence. Evidence from P2-P5 comes from paraphrased notes.

| ID and finding | Severity | Records | Evidence |
|---|---|---|---|
| B1 AI explanations are too dense during fast play | 3 | P1, P2, P3, P5 | P1 (verbatim): "too much info". Paraphrased notes: P2 small AI text; P3 too many words; P5 slightly wordy. Mean T3 ease 2.8/5. |
| B2 Predictions feel like guesses or disconnected text | 2 | P1, P2, P3 | P1 (verbatim): "felt like it was guessing". Paraphrased notes: P2 a guess after quick direction changes; P3 hard to link the text to the boss. |
| B3 Opponent information is slow to locate | 2 | P2, P3, P5 | Paraphrased notes: P2 checked the hero status first; P3 scanned many UI elements; P5 searched the central graphics before the side panel. T2 median 30 s. |
| B4 Recovery: one participant hesitated, and the secondary option was the path used | 1 | P2-P5 | P3 hesitated before recovering (14 s). P2-P5 all recovered with Keep the previous strategy, a text link below the main button (moderator-confirmed). |

## What worked

The entry action was generally easy to find (T1 median 4 s, ease 4.2/5). All five recovered from the preparation failure without help. Graphics and visual design were praised in P1's record and in the notes for P3 and P5; P2's notes mention animation and response time. P4's notes value the detailed AI telemetry and ask for deeper history.

## Observed screen issue X1

The Cyclops error screenshot shows "The rival is not ready" while the disabled start button still says the model is choosing a strategy. This is a researcher observation from a screenshot, not a participant count. It is rated severity 2 and is fixed in the game (Section 4).

The notes for P2, P3 and P5 ask for bolder alerts, icons, short bullets and lane highlights. P4's notes ask for confidence percentages although the panel already showed one, which points to discoverability rather than a missing feature.

# 3 Limitations and data handling

1. P1 T2 and T3 times and outcomes are missing. They are excluded from the corresponding denominators, not treated as zero or failure.
2. P3 T3 outcome is reported as Yes but its notes say prompting was needed. Reported completion is not proof of independent success.
3. Session dates and tested build versions were not recorded. Difficulty consistency and device equivalence are not established.
4. P2's structured gaming frequency is 3-4 times weekly; the note says daily. Both are preserved.
5. T4 ease was recorded for P2-P5 but not P1. It is supplementary and not a consistently administered measure.
6. P1's protocol and error setup evolved during the session. P1's recovery timing boundaries were not separately confirmed.
7. P2-P5's free text was reworded during data entry. It is reported only as paraphrase; none of it is presented as a quote.
8. P2-P5's recorded T4 action named a button that does not exist. The moderator corrected it to Keep the previous strategy. P1's button is unconfirmed.
9. Earlier simulated example records (SIM01-SIM06) and an unrelated P04 fragment are not part of this study and were excluded.

### Interpretation

The five-participant convenience sample is suited to finding interface problems, not to estimating usability or AI accuracy for a population. Task wording evolved during P1, and assistance was not uniform. Phase 5 should keep these distinctions rather than report a uniform independent-success rate.

### No retrospective filling

P1's 40-second runner duration is not used as a T2 or T3 time. A 2/5 ease rating is not converted into a severity score. The missing P1 error count in T4 is not converted to zero. Session dates, build IDs and unknown consent-to-record fields remain absent.

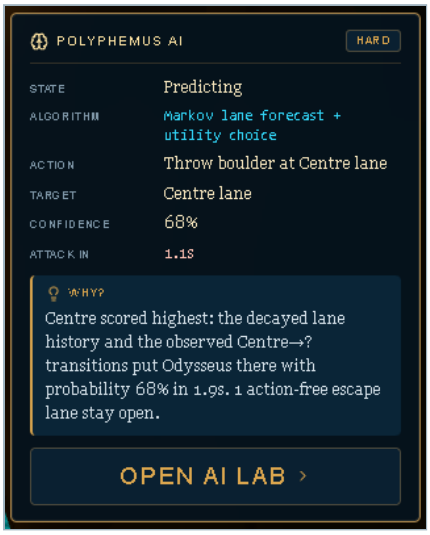
### Next measurement

Phase 5 repeats T1-T4, SUS and the trust question on the fixed build with the same script. Record task boundaries, prompting, device, difficulty and exact build, write down participants' own words at the time, and note which button each participant presses at T4.

# 4 Interface changes from testing

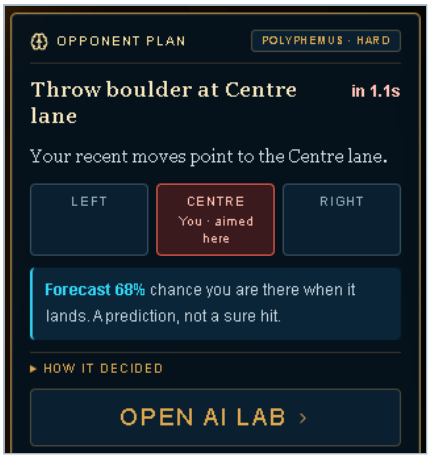
Every change in this section is built into the game, not a separate prototype, and was verified in the browser on November 5, 2026. The before and after panels show the same moment of the same seeded run: adding `?ui=baseline` to the game's URL shows the panel as participants tested it. Captures use the mock AI service.

**BEFORE (AS TESTED)**



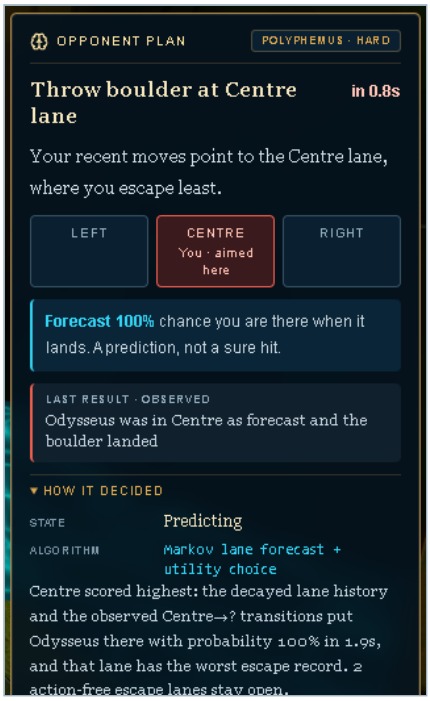

**AI Insights.** State, algorithm, action, target and confidence as equal rows, then a long "Why?" paragraph.

**AFTER**




**Opponent plan.** The action and countdown first, one short reason, a lane cue, and the forecast labelled as a prediction.

**AFTER, LATER IN THE RUN**



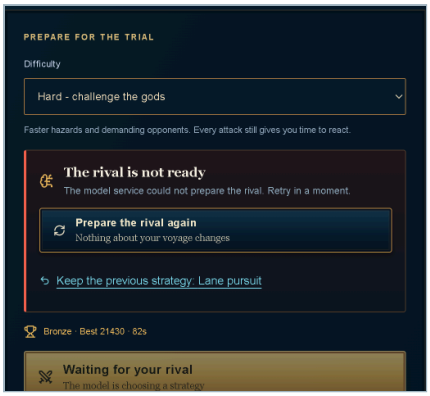

**Last result and details.** The observed result is kept apart from the forecast. "How it decided" opens the algorithm and full reason.

| Finding | Change in the game |
|---|---|
| B1 Dense explanation (severity 3) | The panel leads with the action, then one short reason (e.g. "Your recent moves point to the Centre lane."). The algorithm and the full reasoning sit under "How it decided" and in the AI Lab, so expert detail is kept. |
| B2 Feels like guessing (2) | "Forecast 68% chance you are there when it lands. A prediction, not a sure hit." The last observed result appears under "Last result · observed", and a throw that has landed leaves the headline. |
| B3 Slow to locate (2) | One "Opponent plan" heading. A three-lane cue marks the player's lane ("You") and the lane a locked boulder is aimed at. |

Code: `src/components/ai/ai-inspector.tsx` (panel), `src/lib/ai/cyclops.ts` (short reasons), `src/lib/ai/feeds.ts` (lane cue), `src/styles/farshore-stage.css`.

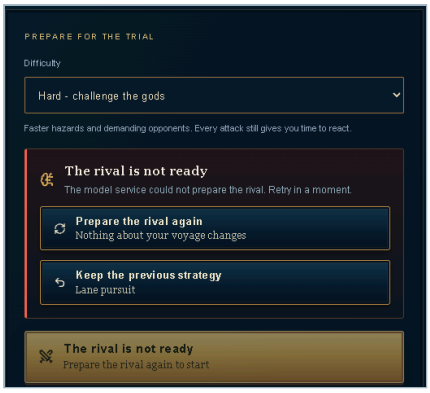
# 4 Recovery states, and remaining issues

**BEFORE (AS TESTED)**



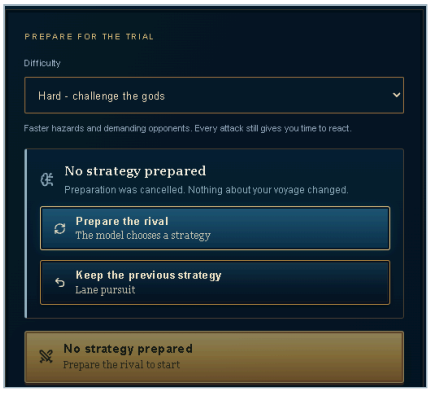

**Error.** Keep the previous strategy is a text link, and the disabled start button says "Waiting for your rival".

**AFTER**




**Error.** Both recovery options are full buttons. The start button names the failure and how to fix it.

**AFTER: NEW STATE**




**Cancelled.** Cancel preparation now stops and waits for the player; before, it restarted at once.

| Finding | Change in the game |
|---|---|
| B4 Recovery (1) | Keep the previous strategy, the option P2-P5 used, is now a full button beside Prepare the rival again, in the error, low-confidence and cancelled states. While loading, Cancel and Keep the previous strategy are both offered. |
| X1 Conflicting labels (2) | The disabled start button names the same state as the panel: "The rival is not ready · Prepare the rival again to start", "No valid strategy yet", "No strategy prepared" or "Waiting for your rival". |

Checks: 268 automated tests pass, including a new one that every Cyclops decision has a short reason of 70 characters or fewer. Scripted browser runs (23 checks at 1440 × 900, 22 at 390 × 844) cover the error, keep-previous, slow, cancel and in-play panel states. Code: `src/ArcadeApp.tsx`, `src/styles/farshore-cinema.css`.

## Remaining issues for Phase 4

| Issue | Plan |
|---|---|
| The fixes have not yet been tested with participants | Phase 5 repeats T1-T4, SUS and trust on the fixed build |
| At phone width the AI panel sits below the play area (M04) | Show a one-line opponent plan inside the play area |
| The Lotus-Eaters maze ignores the model's chosen plan (pilot) | Drive the maze from the chosen plan, then retest |
| The other seven rivals have no short reason yet | Add short reasons; their panels show the full reason meanwhile |
| Long phone briefing (M02), map-label overlap (D05) | Layout fixes |
| The disabled start button keeps its gold look | Mute it while disabled |

# 4. List of changes

Part A lists what changed between the Phase 2 design (the October 8 arcade package, 15 screens and wireframes W01-W28) and the prototype that was tested. Part B lists changes made because of the usability test. Part B is completed only after the sessions, from the Issues sheet in the test data workbook.

Phase 2 personas are referenced as Maya (a gamer who cares about fairness), Daniel (a reader with limited technical comfort) and Rosa (skeptical of AI). They remain the Phase 1 sketches, not new research.

## Part A. Phase 2 design to tested prototype

| ID | Change | Why | Replaces in Phase 2 | Evidence |
|---|---|---|---|---|
| A1 | Five trials became eight: Lotus-Eaters maze rescue, Cyclops' Cave runner, Aeolus wind sailing, Circe's potion duel, Sirens' melody memory, Scylla and Charybdis paddle defense, Calypso's raft hopping, Ithaca archery. | Follows the Odyssey route more closely and gives each episode its own recognizable game. Lotus-Eaters, a maze with simple movement, now comes first. | S03 map with five shores; S05-S09 rounds | `src/lib/arcade/arcade.ts`, commits efcce87-1b8d804 |
| A2 | The AI rival is no longer optional. Every encounter needs a connected model, and there is no model-free mode. | The AI opponent is the core experience, not an add-on. A clear "AI rival offline" state with connection steps replaces the optional toggle. | S04 briefing "mode" choice and A1 optional planning | `src/ArcadeApp.tsx` rival panel; decision recorded 2026-10-26 |
| A3 | The model changed from Llama 3.2 3B to Qwen3-4B-Instruct-2507 (Ollama `qwen3:4b-instruct`). Online accounts can use OpenAI, Groq or OpenRouter instead. | Better instruction and JSON-schema following (MMLU-Pro 69.6 vs 22.2) and no slow thinking phase, which matters for live commentary. Hosted providers help players without a strong laptop. | Phase 2 spike model | README "Connect the AI rival" |
| A4 | Live AI commentary was added: a Homeric bard and the rival comment on play, and the bard writes a short verse after each attempt. Lines are spoken with Kokoro-82M neural voices, with the browser voice as the fallback. | Makes the AI's presence felt during play rather than only before it. Players can switch commentary and voice off separately in Settings. | Not in Phase 2 (model was used only before a round) | commit fbf5cf2; `src/lib/arcade/commentary` |

| ID | Change | Why | Replaces in Phase 2 | Evidence |
|---|---|---|---|---|
| A5 | Explainable AI inspector: AI Insights (default) and AI Lab views over Cyclops' Cave, cycled with F3 or on-screen chips. They show the rival's state, scored candidates, the chosen action, a grounded reason and a fairness guarantee. | Directly answers the Phase 2 trust question: players (Maya, Rosa) can see why the rival acted and that it cannot make an unfair attack. | Phase 2 showed only the plan name before a round | commits d6db373, f9067cd; `src/components/ai/ai-inspector.tsx` |
| A6 | AI Lab (Professor) mode from the main menu runs any encounter at any difficulty, side by side, without touching the campaign save. | Lets an instructor or a curious player compare rival behavior. Supports the class demonstration. | No Phase 2 equivalent | commit d2d0301 |
| A7 | Gameplay moved from a canvas inside the page to a full-viewport 16:9 stage, with F for fullscreen and Escape to pause, and HUD overlays. | The highest-priority item in the redesign brief: the small embedded canvas competed with page chrome for space. | S05-S09 layout, S13 pause | `src/components/play/game-stage.tsx`, `docs/UI_DESIGN.md` |
| A8 | Cyclops' Cave became a 3D three-lane runner (three.js), and Ithaca became a 3D great hall for the archery contest. A 2D fallback is kept for devices without WebGL. | Stronger sense of place and clearer depth cues for dodging and aiming. | S05 grid stealth, S09 archery | commits 4090a08, 14ae287 |
| A9 | Online accounts: email/password and Google sign-in, MongoDB saves across devices, and private per-account AI provider settings. Guests still save on the device. | Lets a signed-in player continue on another device and use a hosted model without a strong laptop. Provider keys stay server-side. | Phase 2 browser-local save only | commits efcce87, 4de1cfe; `SECURITY.md` |
| A10 | Sound rebuilt: per-scene ambience and reverb, and physical-model sound effects that follow game events. Music dips while a voice speaks. | Audio cues reinforce on-screen warnings without replacing them. Every cue is also visible. | Phase 2 optional simple tones | commit 3981e36; `src/lib/audio/` |
| A11 | Briefing redesign: objective, controls, rival portrait and quote, "In Homer / in this game", "How does the AI rival work?", difficulty with a plain-language note, the rival panel and a short practice that saves nothing. | Keeps Daniel's need for a clear objective and practice, and states plainly what the AI decides and what the game decides. | W04-W09 briefing wireframes | `src/ArcadeApp.tsx` |

| ID | Change | Why | Replaces in Phase 2 | Evidence |
|---|---|---|---|---|
| A12 | The voyage map gained a trial gallery and an illustrated voyage chart, with a per-trial AI technique note. | Faster trial choice, and the AI technique is shown before the player commits. | S03 map | `src/components/menu/` |
| A13 | Settings now group AI rival status (models in use, check connection, how to connect), commentary and voice, sound, motion and visual effects, earlier modes, and a showcase "Unlock all islands". | One place to understand and control the AI and accessibility options. | S12 settings overlay | `src/ArcadeApp.tsx` settings dialog |
| A14 | A mock AI service sits behind the single module that makes every model call. It can return success, a slow response, a low-confidence result, an error or "offline", switched from a second browser tab during a session. | The brief asks for every AI state to be shown and tested on demand. Task 4 uses it to lead participants into an AI error without warning. | Phase 2 could show failure states only by stopping the model | `server/lib/mock-ai.ts`, `app/api/ai-mock/route.ts`, `tests/mock-ai.test.mjs` |
| A15 | A separate low-confidence state, "The rival's plan was set aside". It is amber, not red, and its main action is to ask for another strategy. | Before this, a strategy that failed the game's checks looked the same as a connection error. Rosa needs to see that the game refused to use a doubtful result. | Phase 2 A1 error state only | `src/ArcadeApp.tsx` rival panel, frame D22 |
| A16 | Undo for the AI strategy. After "Ask for another strategy", the player can go back to the previous one. While a new one loads, fails or is set aside, "Keep the previous strategy" restores it. | The brief asks users to be able to edit, reject, regenerate or undo AI results. Regenerating used to discard a strategy the player might have preferred. | Phase 2 "ask again" only | `src/ArcadeApp.tsx`, frames D24, D25 |
| A17 | Design tokens: one file defines the palette (background, surface, text, muted text, one accent, an AI colour, error, success), four font families, five text sizes and one spacing scale. The older stylesheets now read them. | Style basics written down once and reused on every screen and in code, as the brief requires. | Colours spread across several stylesheets | `src/styles/tokens.css`, `docs/UI_DESIGN.md` |

# 5 AI tool use statement

AI tools were used for development, analysis and documentation. All participant metrics come from the five session records. No participant, task time, rating, SUS answer or quote was generated to fill a missing field.

| Tool | Used for |
|---|---|
| Claude Code (Anthropic) | Game development and documentation in earlier phases, including the design frames in Section 1 and the Part A change list. On November 5, 2026: built the B1-B4 and X1 fixes into the game, found and fixed the Cancel preparation restart, verified the changes with automated tests and a scripted browser, applied the moderator's data corrections to the workbook, and updated this report and the slides. |
| ChatGPT and Codex (OpenAI) | Inspecting the supplied files, consistency checks, calculations, drafting the first version of Sections 2 and 3, the slides and the workbook, and a standalone HTML prototype of the fixes. While the P2-P5 records were typed up, the free-text notes were reworded; Section 3 treats them as paraphrase only. |
| OpenAI image generation | Game artwork, in earlier phases. |

## What stayed human

The moderator ran the sessions with real participants and recorded the measured values, then confirmed which values stand as recorded and corrected the T4 recovery action.

## Checks

SUS scores were recalculated from all ten answers and match every record. The workbook keeps its formulas and the recorded JSON wording in RawNotes, and its corrections are marked where they apply. The slides and this report use the same numbers.

# 6 Team contribution statement

Farshore | CMPS 4200 HCI | Phase 3
Student and project owner: Bibek Pathak | Solo project

| Member | Contribution |
|---|---|
| Bibek Pathak | Owner of the Farshore project: design direction, implementation decisions and documentation. Ran the five Cyclops' Cave usability sessions as moderator, supplied the session records and the data corrections, and directed the preparation of the report, workbook and slides. |
| AI assistance | Claude Code, ChatGPT and Codex, as disclosed in Section 5. AI tools are not team members or participants. |

The interface fixes from testing are built into the game. Testing them with participants is planned for Phase 5.

Agreed by Bibek Pathak: _____

Date: _____